Analytical Generalized Impulse Responses and Connectedness in Score-Driven Models*

Emilija Dzuverovic Dario Palumbo

Department of Economics, Ca' Foscari University of Venice

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Abstract

We derive analytical generalized impulse responses and construct connectedness measures for a nonlinear score-driven model for strictly positive variables. The model is based on the flexible multivariate generalized beta of the second kind (MGB2) distribution, which accommodates skewness and heavy tails while encompassing several familiar distributions as special or limiting cases. We establish conditions governing shock transmission across series and horizons and explain how market conditions affect connectedness through baseline forecasts. Simulations demonstrate that similar levels of cumulative connectedness can conceal substantially different patterns of dynamic transmission. We apply our framework to daily realized variances of exchange rates and futures. For exchange rates, total connectedness including impact and the following ten trading days is about 40% and nearly constant over time, while impact responses are 2.5 to 4.5 times as large on a selected high-volatility date as on a selected low-volatility date. In the application to four futures series, over the ten trading days after the shock, we identify the S&P 500 E-mini as the largest net transmitter of volatility, followed by crude oil, while natural gas and gold are net receivers. The empirical results underscore the importance of reporting response magnitudes alongside connectedness measures and distinguishing responses on impact from those at later horizons.

Keywords: Generalized impulse response, Connectedness, Score-driven model, MGB2 distribution, Volatility spillovers.

JEL classification: C32, C46, C58

*Dario Palumbo developed the score-driven MGB2 model and proposed extending the connectedness framework to this setting. Emilija Dzuverovic derived the analytical generalized impulse responses and their implications for shock transmission and connectedness, and carried out the simulations and empirical applications.

1 Introduction

Connectedness measures summarize how strongly markets respond to shocks that arise in other markets (Diebold and Yilmaz, 2009, 2012, 2014); see Diebold and Yilmaz (2023) for a review. They reflect relative contributions of shocks, rather than absolute response magnitudes. Similar values of connectedness measures can therefore be associated with substantially different response magnitudes, while cumulative measures can obscure whether responses across markets occur immediately or persist at later horizons. Understanding these distinctions requires a framework that links connectedness measures to the underlying impulse responses. In this paper, we derive analytical generalized impulse responses for a score-driven model for strictly positive variables, such as realized volatility, and use them to construct the corresponding connectedness measures.

Our main result separates the role of market conditions from the relative effect of a shock; see Proposition 1. Market conditions affect the magnitude of a response through the baseline forecast, while the relative response to a shock does not depend on the filtered history. We also establish conditions under which a shock to one series has no effect on another at a given horizon. The dynamics of the model are driven by the score of the multivariate generalized beta of the second kind (MGB2) conditional density (Creal et al., 2013; Harvey, 2013; Palumbo, 2026), and we define generalized shocks as in Koop et al. (1996) and Pesaran and Shin (1998), implying that the responses preserve contemporaneous dependence and do not depend on the ordering of the variables. The responses are tractable because the score is bounded and its moment generating function (MGF) can be evaluated numerically. Connectedness measures are then constructed from these responses by averaging over simulated shocks, following Lanne and Nyberg (2016).

Our results have two implications for the interpretation of connectedness. First, with fixed shape parameters of the MGB2 distribution, connectedness on impact is constant over time and reflects contemporaneous dependence between innovations. Second, a shock can directly affect another series through the score coefficients. The resulting response can persist through the target series' own dynamics. Transmission through intermediate series, however, requires off-diagonal persistence dynamics in our MGB2 specification.

Controlled simulations show why these distinctions matter. In our experiments, total connectedness including impact responses differs by only 1.5 percentage points between a system without and a system with dynamic shock transmission across series. Excluding impact responses, the corresponding values are 0% and 32.4%. Cumulative totals can therefore be similar even when the transmission channels differ strongly.

In the empirical application, we apply our framework to two datasets. First, we show that for daily realized variances of EUR/USD and GBP/USD from 2012 to 2026, total connectedness over $h = 0, \dots, 10$ is about 40% and nearly constant over time. However, on a selected high-volatility date, the impact responses to a shock at the same innovation quantile are 2.5 to 4.5 times as large as on a selected low-volatility date. The VAR benchmark in log realized variances implies lower connectedness, but the difference becomes much smaller when its responses are expressed in realized variance levels and the Lanne–Nyberg generalized forecast error variance decomposition (GFEVD) formula is applied. Connectedness estimates therefore depend on the scale of the responses and the decomposition used, as well as on the fitted dynamics. In the second application, covering four futures series, we identify the S&P 500 E-mini as the largest net transmitter of volatility, followed by crude oil, while natural gas and gold are net receivers. The VAR benchmark implies the same ordering over $h = 0, \dots, 10$ using the Lanne–Nyberg formula, as well as at an 11-day forecast horizon using the conventional GFEVD (Diebold and Yilmaz, 2012, 2014).

We build our framework on generalized impulse responses for nonlinear models (Koop et al., 1996; Pesaran and Shin, 1998), the GFEVD of Lanne and Nyberg (2016), and the connectedness measures of Diebold and Yilmaz (2012). Score-driven models for strictly positive variables (Harvey, 2013; Harvey and Palumbo, 2023; Palumbo, 2026) provide the model class. The importance of the persistence of connectedness is emphasized by Baruník and Křehlík (2018), who develop a frequency-domain decomposition based on the spectral representation of variance decompositions.

Our main contributions are twofold. First, we derive analytical generalized impulse responses for a nonlinear score-driven model based on the flexible MGB2 distribution that accommodates skewness and heavy tails. It encompasses the multivariate Burr and inverted Dirichlet distributions as special cases, while its GB2 marginals include the Burr, Pareto type II, and log-logistic distributions. Generalized gamma and log-normal distributions are obtained as limiting cases. Second, we show how filtered histories affect connectedness through the relative weights assigned to different horizons.

The framework offers several advantages for connectedness analysis. Relative to simulation-based implementations of nonlinear generalized impulse responses, our approach avoids simulating future observations to evaluate the responses. Compared with a Gaussian VAR benchmark in logarithms, we model strictly positive variables directly through nonlinear score-driven dynamics. Therefore, we combine distributional flexibility with analytical tractability, directly explaining how model dynamics and market conditions affect connectedness.

The remainder of the paper is organized as follows. Section 2 introduces the model, the generalized shocks, the analytical responses, and the connectedness measures. Section 3 reports the simulation results, and Section 4 presents the empirical applications. Section 5 discusses limitations and extensions. Proofs, derivations, numerical checks, and additional results are collected in the appendices.

2 Methodology

2.1 Score-driven MGB2 model

Let $y_t = (y_{1t}, \dots, y_{Nt})'$ denote a vector of strictly positive series observed at time t. Following Palumbo (2026), we consider the score-driven multivariate GB2 (MGB2) model for a vector of scales $\alpha_{t|t-1}$, i.e.,

$$y_t = D_t \varepsilon_t, \qquad D_t = \operatorname{diag}(\alpha_{1,t|t-1}, \dots, \alpha_{N,t|t-1}), \qquad \alpha_{i,t|t-1} = \exp(\lambda_{i,t|t-1}), \tag{1}$$

where $N \times 1$ innovation vectors ε_t are i.i.d. and follow an MGB2 distribution with shape parameters $\xi = (\xi_1, \dots, \xi_N)'$, $\nu = (\nu_1, \dots, \nu_N)'$, and ς. The multivariate Burr distribution can be expressed in terms of the MGB2 with $\xi_i = 1$, $\forall i$. The MGB2 specification and its Dirichlet representation follow Cockriel and McDonald (2018) and Sarabia et al. (2021).

Let $\mathcal{F}_{t-1}$ denote the information available at time $t-1$. The score-driven dynamics for $\lambda_{t|t-1}$ are specified as

$$\lambda_{t+1|t} = (I - \Phi_\lambda)\omega_\lambda + \Phi_\lambda \lambda_{t|t-1} + K_\lambda u_t, \tag{2}$$

where $\omega_\lambda \in \mathbb{R}^N$ and $\Phi_\lambda, K_\lambda \in \mathbb{R}^{N \times N}$. A spectral radius of Φ_λ strictly less than one implies that the recursion admits a stationary solution for $\lambda_{t|t-1}$ with mean ω_λ; see Palumbo (2026) for stationarity and invertibility conditions of the MGB2 filter, and also Creal et al. (2013) and Harvey (2013).

The score vector has a convenient Dirichlet representation. Define

$$b_{it} = \frac{(y_{it}/\alpha_{i,t|t-1})^{\nu_i}}{1 + \sum_{j=1}^{N} (y_{jt}/\alpha_{j,t|t-1})^{\nu_j}}, \qquad b_{0t} = \frac{1}{1 + \sum_{j=1}^{N} (y_{jt}/\alpha_{j,t|t-1})^{\nu_j}}, \tag{3}$$

so that $b_{it} = \varepsilon_{it}^{\nu_i} b_{0t}$. Let $b_t = (b_{1t}, \ldots, b_{Nt})'$ and $c = \varsigma + \sum_{i=1}^{N} \xi_i$. Under the MGB2 model, $(b_{1t}, \ldots, b_{Nt}, b_{0t})$ has a Dirichlet distribution with parameters $(\xi_1, \ldots, \xi_N, \varsigma)$, and the score $u_t = \partial \log p(y_t \mid \mathcal{F}_{t-1})/\partial \lambda_{t|t-1}$ is

$$u_t = D_\nu(c\, b_t - \xi), \qquad D_\nu = \operatorname{diag}(\nu_1, \ldots, \nu_N). \tag{4}$$

Hence, $\mathbb{E}(u_t \mid \mathcal{F}_{t-1}) = 0$. Since $b_{it} \in [0, 1]$, the score is bounded, i.e., $u_{it} \in [-\nu_i \xi_i,\ \nu_i(c - \xi_i)]$. This boundedness helps in verifying the conditions for filter invertibility and the asymptotic properties of the maximum likelihood estimator (Blasques et al., 2018). The parameters are estimated by maximizing the conditional MGB2 log-likelihood, while the existence of moments $\mathbb{E}\big[\prod_{i=1}^{N} y_{it}^{m_i} \mid \mathcal{F}_{t-1}\big]$, $m = (m_1, \ldots, m_N)'$, requires $m_i > -\nu_i \xi_i\ \forall i$, and $\sum_i m_i/\nu_i < \varsigma$; see (31) and (38).

2.2 Definition of a generalized shock

Let us consider the shock $z_{it} = \delta_i$, where

$$z_{it} = \varepsilon_{it} - \mu_i, \tag{5}$$

$$\mu_i = \mathbb{E}(\varepsilon_{it}) = \frac{\Gamma(\xi_i + 1/\nu_i)\,\Gamma(\varsigma - 1/\nu_i)}{\Gamma(\xi_i)\,\Gamma(\varsigma)}, \qquad \varsigma > 1/\nu_i,\ \forall i = 1, \ldots, N, \tag{6}$$

equivalently $\varepsilon_{it} = \mu_i + \delta_i > 0$.

We follow the idea initially discussed by Pesaran and Shin (1998), i.e., their impulse response conditions on a single innovation and averages over the remaining innovations according to their conditional distribution. In the linear Gaussian model, this conditional expectation has a closed-form expression. Here, it can be evaluated by simulation from the conditional MGB2 distribution in (12), and Section 2.4 derives it analytically. Unlike a Cholesky transformation, which orthogonalizes innovations using a selected ordering and generally makes the resulting responses order-dependent, conditioning on a single shock leaves the contemporaneously dependent MGB2 innovation vector intact and is invariant to reordering the series. The resulting shock is a reduced-form generalized shock. It is not interpreted as an independently identified structural intervention.

The conditional distribution of the remaining innovation components can be simulated directly from the Dirichlet representation. For the imposed shock δ_i, we set

$$q_i = \frac{(\mu_i + \delta_i)^{\nu_i}}{1 + (\mu_i + \delta_i)^{\nu_i}}, \qquad S_{it} = b_{it} + b_{0t}. \tag{7}$$

Since $b_{it}/b_{0t} = \varepsilon_{it}^{\nu_i}$, conditioning on $\varepsilon_{it} = \mu_i + \delta_i$ fixes $b_{it}/(b_{it} + b_{0t}) = q_i$. The Dirichlet aggregation property gives

$$S_{it} \sim \text{Beta}\Big(\xi_i + \varsigma,\ \sum_{j \neq i} \xi_j\Big), \tag{8}$$

$$(v_{jt})_{j \neq i} \sim \text{Dirichlet}\big((\xi_j)_{j \neq i}\big), \qquad S_{it} \perp (v_{jt})_{j \neq i}, \tag{9}$$

$$b_{it} = q_i S_{it}, \qquad b_{0t} = (1 - q_i) S_{it}, \tag{10}$$

$$b_{jt} = (1 - S_{it}) v_{jt}, \qquad j \neq i, \tag{11}$$

$$\varepsilon_{jt} = \left(\frac{b_{jt}}{b_{0t}}\right)^{1/\nu_j}, \qquad j \neq i. \tag{12}$$

This sampler preserves the correct contemporaneous dependence among the components when a shock to one series is imposed. It also gives the full score vector through (4), which is then used to update (2).

2.3 Generalized impulse responses

Following Koop et al. (1996) and Pesaran and Shin (1998), we define the response to a shock δ_i in series i, conditional on $\mathcal{F}_{t-1}$, as

$$\mathrm{GIRF}_i(h;\delta_i,\mathcal{F}_{t-1}) = \mathbb{E}\big[y_{t+h} \mid \mathcal{F}_{t-1}, z_{it}=\delta_i\big] - \mathbb{E}\big[y_{t+h} \mid \mathcal{F}_{t-1}\big], \qquad h=0,1,\ldots,H. \tag{13}$$

This response lies in $\mathbb{R}^N$. Let e_j be the selection vector with the jth element equal to one and zeros elsewhere. The response of series j to i, used in the variance decomposition below, is the jth component

$$\mathrm{GIRF}_{j\leftarrow i}(h;\delta_i,\mathcal{F}_{t-1}) = e_j' \,\mathrm{GIRF}_i(h;\delta_i,\mathcal{F}_{t-1}).$$

The first expectation in (13) conditions on the value of the ith innovation and integrates over the conditional MGB2 distribution of the other components. The second expectation is the standard MGB2 forecast given the information set $\mathcal{F}_{t-1}$. In particular, iterating forward (2) and using $\mathbb{E}(u_t \mid \mathcal{F}_{t-1}) = 0$ gives

$$\mathbb{E}\big[\lambda_{t+h|t+h-1} \mid \mathcal{F}_{t-1}\big] = \omega_\lambda + \Phi_\lambda^h(\lambda_{t|t-1} - \omega_\lambda), \qquad h \geq 0. \tag{14}$$

Let us define

$$a_{j,t}(h) = e_j'\big[\omega_\lambda + \Phi_\lambda^h(\lambda_{t|t-1} - \omega_\lambda)\big], \qquad s_j(r) = K_\lambda'(\Phi_\lambda')^r e_j. \tag{15}$$

Using the linear filter representation and the Dirichlet moment generating function (MGF), we can show

$$\mathbb{E}\big[\alpha_{j,t+h|t+h-1} \mid \mathcal{F}_{t-1}\big] = \exp(a_{j,t}(h)) \prod_{r=0}^{h-1} M_u(s_j(r)), \tag{16}$$

$$\mathbb{E}\big[y_{j,t+h} \mid \mathcal{F}_{t-1}\big] = \mu_j \exp(a_{j,t}(h)) \prod_{r=0}^{h-1} M_u(s_j(r)), \tag{17}$$

where

$$M_u(s) = \mathbb{E}\big[\exp(s'u_t)\big] = \exp(-s'D_\nu\xi) \sum_{k\in\mathbb{N}_0^N} \frac{\Gamma(c)}{\Gamma\big(c+\sum_{l=1}^N k_l\big)} \prod_{l=1}^N \frac{\Gamma(\xi_l+k_l)\,(c\nu_l s_l)^{k_l}}{\Gamma(\xi_l)\,k_l!}. \tag{18}$$

Since b_t is bounded, $M_u(s)$ is finite for all $s \in \mathbb{R}^N$.

Appendix A presents the derivation of the baseline forecasts. The first expectation in (13) is derived in Section 2.4, where it is expressed analytically in terms of MGFs that are evaluated numerically. At later horizons, any effect of the shock propagates through the recursion (2) via u_t, while future scores enter through the unconditional MGF in (18).

2.4 Analytical generalized impulse responses

The Dirichlet representation used in Section 2.2 implies that the response in (13) admits an exact analytical expression given the following independence result.

Lemma 1. *In the score-driven MGB2 model, for any source series i, the vector $b_{-i,t} = (b_{jt})_{j\neq i}$ is independent of ε_{it}. Consequently, $u_{-i,t} = (u_{jt})_{j\neq i}$ is independent of ε_{it}, and conditioning on $z_{it} = \delta_i$ leaves the joint distribution of $u_{-i,t}$ unchanged. In particular, $\mathbb{E}(u_{jt} \mid \mathcal{F}_{t-1}, z_{it}=\delta_i) = 0$ $\forall j \neq i$.*

The other innovations ε_{jt}, $j \neq i$, respond to the shock because they share the common gamma variable G_{0t} in (30). For $j \neq i$,

$$\mathbb{E}\big(\varepsilon_{jt} \mid \varepsilon_{it} = x\big) = (1+x^{\nu_i})^{1/\nu_j} \,\frac{\Gamma(\xi_j+1/\nu_j)\,\Gamma(\xi_i+\varsigma-1/\nu_j)}{\Gamma(\xi_j)\,\Gamma(\xi_i+\varsigma)}, \qquad \xi_i+\varsigma > 1/\nu_j, \tag{19}$$

which is increasing in x. Let us define the conditional MGF of the score

$$M_u^{(i)}(s;q) = \mathbb{E}\big[\exp(s'u_t) \mid q_{it} = q\big] = \exp(-s'D_\nu\xi)\,\mathbb{E}\Big[\exp\Big\{c\Big(\nu_i s_i\, q\, S_{it} + (1-S_{it})\sum_{k\neq i}\nu_k s_k v_{kt}\Big)\Big\}\Big], \tag{20}$$

where $q_{it} = b_{it}/(b_{it}+b_{0t})$ and $(S_{it}, v_{-i,t})$ are distributed as in (8)–(9). Conditional on S_{it}, the inner expectation is the MGF of the Dirichlet vector $v_{-i,t}$. As in (34), it can be expanded in a series using the Dirichlet moments (33), here with parameters $(\xi_k)_{k\neq i}$.

Proposition 1. *Let $m_{j,t}(h) = \mathbb{E}[y_{j,t+h} \mid \mathcal{F}_{t-1}]$ be the baseline forecast (17), and let q_i be given by (7). Then, for every $h \geq 0$,*

$$\mathrm{GIRF}_{j\leftarrow i}(h;\delta_i,\mathcal{F}_{t-1}) = m_{j,t}(h)\, g_{ji}(h;\delta_i), \tag{21}$$

where g_{ji} does not depend on $\mathcal{F}_{t-1}$,

$$g_{ii}(0;\delta_i) = \delta_i/\mu_i, \tag{22}$$

$$g_{ji}(0;\delta_i) = \mathbb{E}\big(\varepsilon_{jt} \mid \varepsilon_{it} = \mu_i + \delta_i\big)/\mu_j - 1, \qquad j \neq i, \tag{23}$$

$$g_{ji}(h;\delta_i) = \frac{M_u^{(i)}\big(s_j(h-1);q_i\big)}{M_u\big(s_j(h-1)\big)} - 1, \qquad h \geq 1. \tag{24}$$

Corollary 1. *Under Proposition 1, let $h \geq 1$ and $j \neq i$. If*

$$(\Phi_\lambda^{h-1}K_\lambda)_{ji} = 0,$$

then $g_{ji}(h;\delta_i) = 0$, and hence $\mathrm{GIRF}_{j\leftarrow i}(h;\delta_i,\mathcal{F}_{t-1}) = 0$, for every admissible shock size δ_i and every filtered history $\mathcal{F}_{t-1}$.

With diagonal Φ_λ, the score u_{it} enters the response index of series j with coefficient $\phi_j^{h-1}K_{ji}$ at $h \geq 1$. Thus, a nonzero K_{ji} can generate a direct response that persists through series j's own dynamics. However, diagonal Φ_λ does not pass the original shock along a chain of score coefficients, i.e., if $K_{31} = 0$, then $K_{21} \neq 0$ and $K_{32} \neq 0$ do not produce a response of series 3 to a shock in series 1 at $h \geq 1$. Off-diagonal elements of Φ_λ can transmit the effect along such a chain. Section 3 illustrates both cases.

Proofs are given in Appendix B, while Appendix C provides a Monte Carlo assessment of the analytical responses.

2.5 Generalized forecast error variance decomposition (GFEVD)

We construct generalized forecast error variance decompositions (GFEVD) from impulse responses in (13), following Lanne and Nyberg (2016). Given $\mu = \mathbb{E}(\varepsilon_t)$, let us draw a vector $\varepsilon^{(b)}$ from the fitted MGB2 distribution. The corresponding vector of shocks is $\delta^{(b)} = \varepsilon^{(b)} - \mu$.

For a given filtered history $\mathcal{F}_{t-1}$ and shock vector δ, the response to source i conditions only on $z_{it} = \delta_i$. The other components of the time t innovation vector are integrated out under their conditional MGB2 distribution, as described in (12), i.e., they are not fixed at the other entries of δ. For target series j, source series i, and horizon H, let us define

$$\lambda_{ji,t}^{(H)}(\delta) = \frac{\sum_{h=0}^{H}\mathrm{GIRF}_{j\leftarrow i}(h;\delta_i,\mathcal{F}_{t-1})^2}{\sum_{r=1}^{N}\sum_{h=0}^{H}\mathrm{GIRF}_{j\leftarrow r}(h;\delta_r,\mathcal{F}_{t-1})^2}, \tag{25}$$

provided the denominator is positive. The numerator measures the cumulative squared response of series j to source i, while the denominator sums the corresponding responses over all sources. Consequently, $\sum_{i=1}^{N}\lambda_{ji,t}^{(H)}(\delta) = 1$ for every time period and shock vector.

Substituting (21) into (25) gives

$$\lambda_{ji,t}^{(H)}(\delta) = \frac{\sum_{h=0}^{H} m_{j,t}(h)^2 \, g_{ji}(h;\delta_i)^2}{\sum_{r=1}^{N}\sum_{h=0}^{H} m_{j,t}(h)^2 \, g_{jr}(h;\delta_r)^2}. \tag{26}$$

Since the same forecast path $m_{j,t}(h)$ appears in the numerator and the denominator, the shares are unaffected by the scale of the target series. Moreover, the filtered history affects the shares only through the relative forecast path $m_{j,t}(h)/m_{j,t}(0)$, $h = 0, \ldots, H$. Since

$$\frac{m_{j,t}(h)}{m_{j,t}(0)} = \exp\left\{e_j'(\Phi_\lambda^h - I)(\lambda_{t|t-1} - \omega_\lambda)\right\} \prod_{r=0}^{h-1} M_u(s_j(r)),$$

time variation in the shares is driven by the distance of $\lambda_{t|t-1}$ from ω_λ and by the speed of mean reversion. Two implications follow.

Corollary 2. *For $H = 0$, $\lambda_{ji,t}^{(0)}(\delta) = g_{ji}(0;\delta_i)^2 / \sum_r g_{jr}(0;\delta_r)^2$ does not depend on t. The impact connectedness table is therefore constant over time and measures purely contemporaneous dependence, which operates through the common component governed by ς.*

Corollary 3. *If Φ_λ and K_λ are diagonal (Appendix A.3), then $g_{ji}(h;\delta_i) = 0$ for all $h \geq 1$ and $j \neq i$. Spillovers are then purely contemporaneous, and the off-diagonal shares are nonincreasing as H increases. Dynamic transmission requires non-zero off-diagonal elements of K_λ or Φ_λ.*

The time t generalized share averages the above normalized contributions over joint draws of shocks, i.e.,

$$\theta_{ji,t}^{(H)} = \mathbb{E}_\delta\left[\lambda_{ji,t}^{(H)}(\delta) \mid \mathcal{F}_{t-1}\right], \qquad \sum_{i=1}^{N} \theta_{ji,t}^{(H)} = 1. \tag{27}$$

In computation, we evaluate (25) for each draw $\delta^{(b)}$ and then average the resulting shares. Drawing from the fitted MGB2 distribution is the parametric counterpart of the residual bootstrap proposed by Lanne and Nyberg (2016). Averaging $\theta_{ji,t}^{(H)}$ over filtered histories gives a sample average connectedness table. Decompositions can also be reported for individual time periods or for selected sets of histories and shock sizes.

To examine shock propagation after impact, we compute shares over $h = 1, \ldots, H$ by replacing the horizon sums in (25) with sums starting at $h = 1$ and applying the same averaging procedure, provided that the denominators are positive. These shares are normalized separately from those including impact responses.

We summarize the generalized shares obtained using the Lanne–Nyberg criterion through the connectedness framework of Diebold and Yilmaz (2014). In particular, total connectedness at time t is

$$C_t^{(H)} = \frac{100}{N} \sum_{j=1}^{N} \sum_{i \neq j} \theta_{ji,t}^{(H)}. \tag{28}$$

Directional connectedness received by series j from the other series, and transmitted by series i to the other series, are respectively

$$C_{j\leftarrow\bullet,t}^{(H)} = 100 \sum_{i \neq j} \theta_{ji,t}^{(H)}, \qquad C_{\bullet\leftarrow i,t}^{(H)} = 100 \sum_{j \neq i} \theta_{ji,t}^{(H)}. \tag{29}$$

Net connectedness for series i is $C_{\bullet\leftarrow i,t}^{(H)} - C_{i\leftarrow\bullet,t}^{(H)}$. Pairwise measures follow from the corresponding off-diagonal entries of the same table. By Corollary 2, $\theta_{ji,t}^{(0)}$, and hence $C_t^{(0)}$ and the directional, net and pairwise measures at $H = 0$, do not depend on t; time variation in connectedness arises only for $H > 0$, through the relative forecast paths in (26).

The entries in the connectedness table measure relative contributions to cumulative squared generalized responses, with each row summing to one by construction. They do not necessarily represent separate additive contributions to the conventional forecast error variance.

2.6 Extensions

With fixed shape parameters, the impact connectedness table is constant over time (Corollary 2). Since contemporaneous dependence operates through the common gamma component G_{0t} in (30), whose shape is ς, a natural extension lets ς vary over time, in the spirit of the time-varying shape parameters in the score-driven MGB2 models of Palumbo (2026). For instance, $\varsigma_{t|t-1} = \underline{\varsigma} + \exp(f_{t|t-1})$ with $\underline{\varsigma} = \max_j 1/\nu_j$, where $f_{t|t-1}$ is driven by the score $(\varsigma_{t|t-1} - \underline{\varsigma})[\psi(c_t) - \psi(\varsigma_{t|t-1}) + \log b_{0t}]$, $c_t = \varsigma_{t|t-1} + \sum_i \xi_i$, and ψ is the digamma function. Impact connectedness would then vary over time, and a shock could also affect future contemporaneous dependence through b_{0t}. Several issues would need to be addressed. The lower bound $\underline{\varsigma}$ guarantees finite contemporaneous conditional means only, while the finiteness of multi-step forecasts requires further conditions. The analytical results of Section 2.4 would no longer apply for $h \geq 1$. Finally, the model with fixed shape parameters is recovered when the score coefficient of $f_{t|t-1}$ is zero only under a suitable initialization, e.g., $f_{1|0}$ equal to its unconditional mean. We leave this extension for future research.

3 Simulations

We consider $N = 3$ series to illustrate direct and indirect shock transmission under known parameter values. All responses and GFEVD shares are computed via Proposition 1 at the true parameters, showing what each part of the specification implies for the responses and the connectedness measures.

In all experiments, we set $\xi = (1.5, 2.0, 1.2)'$, $\nu = (1.8, 1.3, 2.2)'$, and $\omega_{\lambda,j} = -\log \mu_j$. Furthermore, $\Phi_0 = \mathrm{diag}(0.90, 0.85, 0.80)$ and $K_0 = \mathrm{diag}(0.05, 0.04, 0.06)$. Let L be an $N \times N$ matrix with $L_{21} = L_{32} = 1$ and zeros elsewhere. The five settings are

Design	Φ_λ	K_λ	ς
A	Φ_0	K_0	3
B	Φ_0	$K_0 + 0.03L$	3
C	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	3
C ($\varsigma = 2$)	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	2
C ($\varsigma = 6$)	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	6

Design A implies diagonal dynamics. Design B introduces direct effects through the scores from series 1 to series 2 and from series 2 to series 3. Design C additionally allows a shock to series 1 to affect series 3 at later horizons. The last two settings use different values of the shape parameter ς, which jointly governs the upper tails, the dependence among the innovations, and the distribution of the score. In every experiment, $\rho(\Phi_\lambda) = 0.90$, ensuring a stationary log-scale recursion, and $\nu_j \varsigma > 2 \ \forall j$, ensuring finite second moments. Given $\varsigma = 2$, $\nu_j \varsigma = (3.6, 2.6, 4.4)$, so the first two series have no finite fourth moment. Responses are computed for $h = 0, \ldots, 20$ at the two initial log-scale vectors, i.e., $\lambda^{(a)} = \omega_\lambda$ and $\lambda^{(b)} = \omega_\lambda + (0.5, -0.4, 0.3)'$, for shocks at the 10th and 90th percentiles of ε_{it}. The GFEVD shares are computed by averaging over $B = 32{,}000$ shock draws.[1]

Figure 1 reports the relative responses $g_{j1}(h; \delta_1)$ to a 90th-percentile shock to the first series (top row) and the generalized impulse responses under Design C at both histories (bottom row). In Design A, the responses of the second and third series vanish at $h \geq 1$, as stated in Corollary 3. In Design B, the score coefficient K_{21} generates a response of the second series from

[1] The MC standard errors of the shares and total connectedness are at most 0.19 percentage points. When comparing histories, we use common shock draws and calculate the standard errors from the draw-by-draw differences. As an independent check, the responses to a 90th-percentile shock to the first series at $\lambda^{(b)}$ were also estimated by simulation without using Proposition 1 in Designs B, C, and C with $\varsigma = 6$, via 4×10^5 paths. The largest absolute difference between the analytical and simulated responses is 2.7×10^{-3}, while in two reruns with 1.6×10^6 paths it falls to at most 1.3×10^{-3}.

$h = 1$ onward. Given diagonal Φ_λ, $(\Phi_\lambda^{h-1} K_\lambda)_{21} = \phi_2^{h-1} K_{21}$, so the weight of the first series' score in the response index of the second series decays geometrically at the rate ϕ_2. Because $K_{31} = 0$, the third series does not respond to the first at any horizon $h \geq 1$, although $K_{21} \neq 0$ and $K_{32} \neq 0$, as implied by Corollary 1. In Design C, $(\Phi_\lambda K_\lambda)_{31} = 0.08 \times 0.03 \neq 0$, so the third series responds to the first with a delay. In particular, the relative response is zero at $h = 1$, peaks at about 2% of the baseline forecast near $h = 10$, and then decays slowly. In addition, the response of the second series to the first is more persistent than in Design B.

The bottom row shows that history affects the generalized impulse response, although the relative responses do not depend on it. At $\lambda^{(b)}$, the first series has a higher baseline forecast and a larger own response. The second series starts with a log-scale below its long-run level, so its response to the same shock is smaller at the early horizons. Subsequently, its response approaches that at $\lambda^{(a)}$.

Table 1 summarizes the GFEVD shares. Impact connectedness ($h = 0$) is the same in Designs A–C because they share the same innovation distribution. It is also history independent, as implied by Corollary 2. For horizons $h = 1, \ldots, 20$, all series respond only to their own shocks in Design A. In Design B, the shares resulting from shocks to the other series occur only for the directly connected pairs, $2 \leftarrow 1$ (48.0%) and $3 \leftarrow 2$ (23.6%). In Design C, the GFEVD share from series 1 to series 3 is 11.1%, although $K_{31} = 0$, and the share associated with $2 \leftarrow 1$ increases to 61.1%. Total connectedness over $h = 0, \ldots, 20$ differs by only 1.5 percentage points between Designs A and C, whereas the total over $h = 1, \ldots, 20$ ranges from zero to 32.4%.

In these experiments, changing the shape parameter ς mainly affects the impact shares. In particular, decreasing ς from 6 to 2 raises total impact connectedness from 24.7% to 45.6%, whereas the totals over $h = 1, \ldots, 20$ remain between 32.3% and 32.6%. With $\varsigma = 6$, the latter exceeds impact connectedness.[2] Figure 8 and Table 8 in Appendix D report the responses for the three values of ς and the full set of GFEVD shares.

Finally, total connectedness over $h = 0, \ldots, 20$ at $\lambda^{(b)}$ in Design A exceeds that at $\lambda^{(a)}$ by 0.23 percentage points because the baseline forecasts change the relative weight of the impact and later horizons. The difference is larger in Design C, i.e., 0.61 percentage points. In this case, history also affects the GFEVD shares over $h = 1, \ldots, 20$.

4 Empirical application

4.1 Application to exchange rates

In the first part of the empirical analysis, we apply the model to daily realized variances of EUR/USD and GBP/USD from the VOLARE database (Cipollini et al., 2026), which computes realized measures from Kibot intraday bid–ask quotes.[3] In particular, we use the series of 5-minute realized variances y_t (i.e., multiplied by 252 and by 100) from January 2, 2012, to the end of September 2026, resulting in 3,825 daily observations.

The time series means (standard deviations) of the realized variances are 0.615 (0.663) for EUR/USD and 0.748 (1.531) for GBP/USD. The larger dispersion of GBP/USD reflects a few extreme observations, and its largest value is recorded on June 23, 2016, around the Brexit referendum. The logarithms of the two series have a correlation of 0.69, indicating substantial comovement. Their first-order autocorrelations are 0.69 and 0.71, respectively, declining to 0.43 and 0.37 at lag 22.

We estimate the model (1)–(4) under the multivariate Burr distribution, where $\xi_i = 1$, $\forall i$, in three versions. The first has diagonal parameter matrices Φ_λ and K_λ. The second consists of diagonal Φ_λ and unrestricted K_λ, while the third specification implies both Φ_λ and K_λ

[2] Reported shares are normalized by their respective totals of squared responses. Their comparison therefore concerns relative contributions rather than absolute response magnitudes.

[3] For exchange rates, the measures are constructed from mid-quote prices aggregated to 1-second frequency.

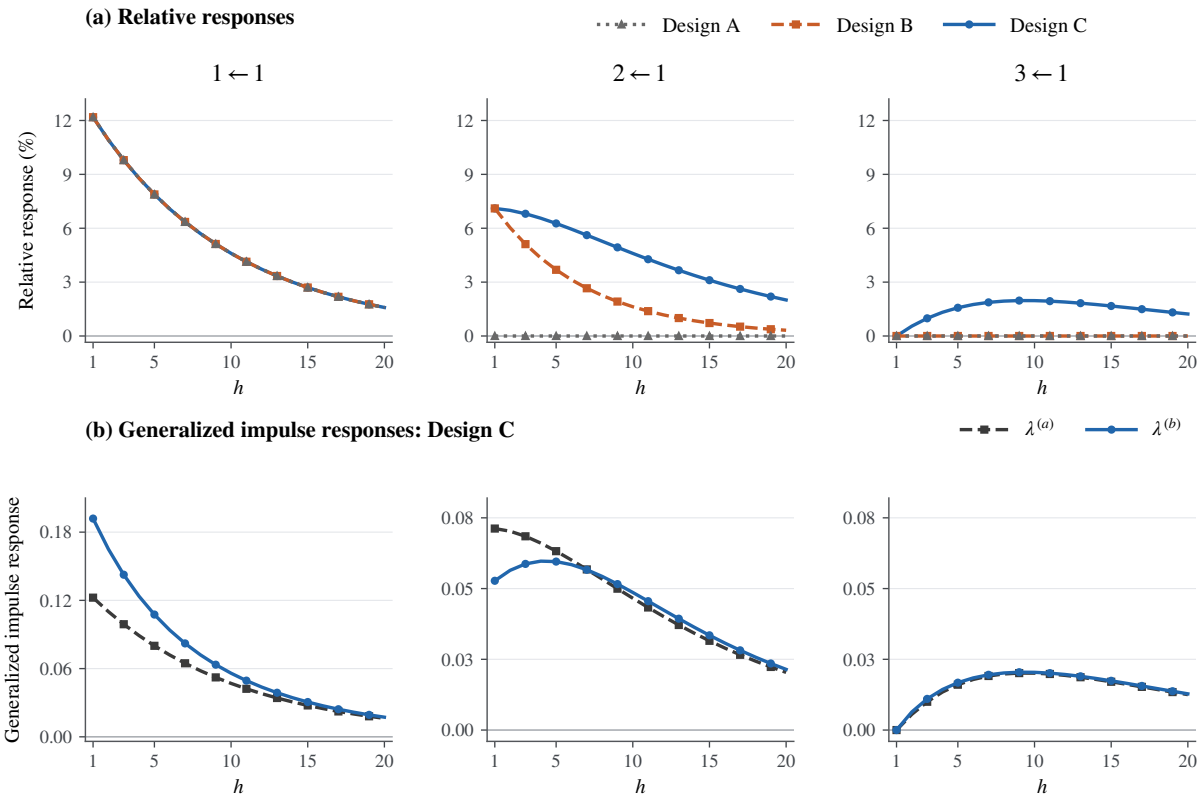


Figure 1: Responses to a 90th-percentile shock to series 1 at horizons $h = 1, \ldots, 20$. Top: relative responses $g_{j1}(h; \delta_1)$ in Designs A–C, expressed as percentages of the baseline forecast. Bottom: generalized impulse responses in Design C at histories $\lambda^{(a)} = \omega_\lambda$ and $\lambda^{(b)} = \omega_\lambda + (0.5, -0.4, 0.3)'$.

Table 1: Connectedness in the controlled settings

	Total			Connectedness over $1 \le h \le 20$				Total $\lambda^{(b)} - \lambda^{(a)}$ (pp)	
Design	$h = 0$	$h \le 20$	$1 \le h \le 20$	$2 \leftarrow 1$	$3 \leftarrow 2$	$3 \leftarrow 1$	$1 \leftarrow 2$	$h \le 20$	$1 \le h \le 20$
A	37.6	36.1	0.0	0.0	0.0	0.0	0.0	+0.23	+0.00
B	37.6	36.8	23.9	48.0	23.6	0.0	0.0	+0.28	−0.01
C	37.6	37.6	32.4	61.1	25.1	11.1	0.0	+0.61	+0.26
C ($\varsigma = 2$)	45.6	44.9	32.3	62.0	24.1	10.8	0.0	+0.47	+0.27
C ($\varsigma = 6$)	24.7	27.0	32.6	60.0	26.4	11.4	0.0	+0.98	+0.24

Notes: The table reports total connectedness and selected Lanne–Nyberg GFEVD shares (in %) computed from Proposition 1 at the true parameters and averaged over $B = 32{,}000$ shock draws. The first seven numerical columns report values evaluated at history $\lambda^{(a)}$; the final two report differences in total connectedness between histories $\lambda^{(b)}$ and $\lambda^{(a)}$, in percentage points. "$h \le 20$" includes horizons $h = 0, \ldots, 20$, while "$1 \le h \le 20$" excludes impact responses and uses a separate normalization. All denominators are positive for the sampled shocks.

unrestricted. To avoid convergence to local maxima, we consider multiple starting values for the parameters. The filter starts at $\lambda_{1|0} = \omega_\lambda$.[4] Table 2 reports the estimation results. Allowing an unrestricted coefficient matrix K_λ increases the log-likelihood by 209.5 points, from 1,223.86 to 1,433.35. Allowing unrestricted Φ_λ raises it by a further 2.4 points. BIC favors the specification with diagonal Φ_λ and unrestricted K_λ, which we use as the baseline, while AIC selects the fully unrestricted specification.[5]

The baseline estimates indicate highly persistent log-scales. Both persistence parameters are around 0.98, implying a half-life of approximately 32 trading days. High persistence is also reported by Harvey and Palumbo (2023), whose two-component score-driven models for major equity indices feature a near unit-root long-run component and a short-run component with persistence between 0.82 and 0.89. Diagonal score coefficients k_{11} and k_{22} are both around 0.05, while the off-diagonal k_{12} and k_{21} are approximately 0.03. Their implications for shock transmission are assessed through the generalized impulse responses below. The fitted tail indices imply finite second moments at the point estimates. In particular, $\nu_{i\varsigma} = (2.48, 2.72)$ in the diagonal specification, $(2.45, 2.68)$ in the baseline, and $(2.46, 2.69)$ in the specification with unrestricted coefficient matrices.

Figure 2 reports relative responses to shocks at the 10th and 90th percentiles of each innovation. These responses are history independent by Proposition 1. A 90th-percentile shock to EUR/USD raises its realized variance by approximately 65% on impact and the expected realized variance of GBP/USD by 48%, relative to their respective baseline forecasts. The corresponding shock to GBP/USD raises its own realized variance by 60% and the expected realized variance of EUR/USD by 51%. At $h = 1$, responses to own shocks fall to about 12% and responses to shocks in the other series to approximately 7–8%. By $h = 20$, they decline to around 8% and 5%, respectively. Responses to 10th-percentile shocks are negative and smaller in absolute value on impact, but larger in absolute value than those of the positive shocks at later horizons.

Figure 3 compares the generalized impulse responses on a relatively low-volatility day (June 11, 2026) and a relatively high-volatility day (December 13, 2018). These dates correspond to the 10th and 90th percentiles, respectively, of the average deviation of the fitted log-scales from their long-run levels. On the high-volatility day, impact responses are 2.5 times as large for EUR/USD and 4.5 times as large for GBP/USD as on the low-volatility day. At $h = 20$, the corresponding ratios are 1.8 and 2.6. A shock at the same innovation quantile therefore produces different changes in realized variances across distinct market conditions.

Table 3 reports sample averages of the GFEVD shares. In general, these are not additive contributions to the conventional forecast error variance. Impact connectedness is history independent by Corollary 2. Both cross-series impact shares are approximately 44% and the net impact measure for EUR/USD is -0.49 percentage points. Over $h = 1, \ldots, 10$, the GFEVD share associated with GBP/USD shocks is 36.83% for EUR/USD, while the reverse share is 34.84%. The resulting net measure for EUR/USD is -1.99 percentage points. Total connectedness is 43.99% on impact, 40.04% cumulatively over $h = 0, \ldots, 10$, and 35.84% over $h = 1, \ldots, 10$.

The directional imbalance is modest and depends on the specification and horizon. With unrestricted Φ_λ, the net measure for EUR/USD is -1.73 percentage points over $h = 1, \ldots, 10$, but changes sign, reaching $+0.41$ over $h = 1, \ldots, 20$, compared with -1.98 in the baseline.

Connectedness varies slightly over time in the baseline specification. Total connectedness over $h = 0, \ldots, 10$ ranges from 39.53% to 40.57%, while the net EUR measure has a range of 0.89 percentage points. Over $h = 1, \ldots, 10$, the ranges of both measures are about 0.02 percentage points (Figure 9 in Appendix D). This near constancy is a feature of the estimates,

[4] To assess the effect of initializing the filter at $\lambda_{1|0} = \omega_\lambda$, we rerun it at the estimates from alternative initial log-scales ($\omega_\lambda \pm 1$ and the log of the average of the first 22 observations). The filtered paths converge after approximately 100 observations in each specification.

[5] We report robust standard errors based on a Newey–West sandwich covariance estimator. Robust Wald tests reject diagonal K_λ in the second specification and jointly diagonal Φ_λ and K_λ in the third at the 5% significance level. In the fully unrestricted specification, diagonal Φ_λ is not rejected at the 5% level.

Table 2: Estimation results

	Diagonal Φ_λ and K_λ	K_λ unrestricted	Both matrices unrestricted
$\log \nu_1$	1.447	1.490	1.489
	(0.024)	(0.023)	(0.023)
$\log \nu_2$	1.542	1.580	1.580
	(0.027)	(0.026)	(0.027)
$\log \varsigma$	−0.540	−0.593	−0.591
	(0.045)	(0.036)	(0.036)
$\omega_{\lambda,1}$	−0.433	−0.957	−0.959
	(0.243)	(0.093)	(0.096)
$\omega_{\lambda,2}$	−0.782	−0.768	−0.769
	(0.101)	(0.101)	(0.089)
ϕ_{11}	0.996	0.979	0.987
	(0.003)	(0.004)	(0.005)
ϕ_{12}	–	–	−0.010
			(0.006)
ϕ_{21}	–	–	0.009
			(0.007)
ϕ_{22}	0.982	0.979	0.969
	(0.005)	(0.004)	(0.008)
k_{11}	0.027	0.049	0.049
	(0.003)	(0.005)	(0.006)
k_{12}	–	0.030	0.031
		(0.004)	(0.004)
k_{21}	–	0.033	0.032
		(0.004)	(0.005)
k_{22}	0.029	0.047	0.047
	(0.003)	(0.006)	(0.006)
$\rho(\Phi_\lambda)$	0.996	0.979	0.978
LLF	1,223.86	1,433.35	1,435.77
Total nb. of parameters	9	11	13
AIC	−2,429.7	−2,844.7	−2,845.5
BIC	−2,373.5	−2,776.0	−2,764.3

Notes: The table reports estimates of the model (1)–(4) under the Burr distribution ($\xi_1 = \xi_2 = 1$) and three dynamic specifications for EUR/USD (1) and GBP/USD (2) realized variances. The sample covers January 2, 2012, to September 30, 2026, containing 3,825 daily observations. Robust Newey–West standard errors are in parentheses. Shape parameters are reported on the log scale.

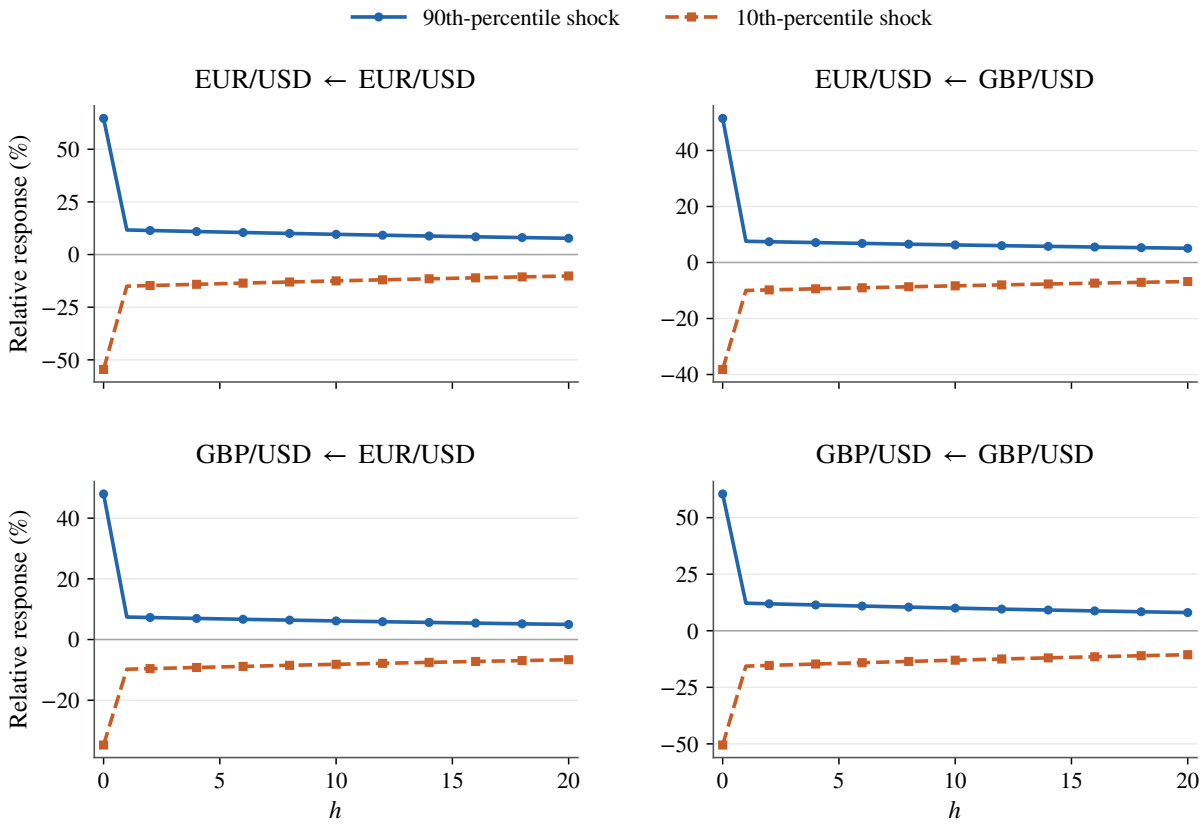


Figure 2: Relative responses $g_{ji}(h;\delta_i)$ to shocks at the 10th and 90th percentiles of each innovation, expressed as percentages of the baseline forecast. Burr baseline specification ($\xi_1 = \xi_2 = 1$, diagonal Φ_λ and unrestricted K_λ), horizons $h = 0, \ldots, 20$.

not a general consequence of diagonal Φ_λ, as the simulations in Section 3 illustrate. The fitted responses are sufficiently similar across source series that the history-dependent forecast weights nearly cancel in the shares. Normalized connectedness can therefore remain stable, while the underlying response magnitudes vary substantially across histories.

As a benchmark[6], we estimate a VAR in log realized variances and construct connectedness measures from its normalized generalized FEVD, following Diebold and Yilmaz (2012, 2014). Estimation is based on the same sample period, with BIC selecting a VAR(10) from candidate orders 1 to 22. The decomposition is computed from the corresponding MA representation. Total connectedness is 32.90% and net EUR connectedness is +0.02 percentage points at an 11-day forecast horizon, matching $h = 0, \ldots, 10$ in our analysis. At a 21-day forecast horizon, matching $h = 0, \ldots, 20$, the corresponding values are 32.74% and +1.30 percentage points.

For a closer comparison, we use the same estimated VAR to compute generalized impulse responses of realized variance in levels under Gaussian innovations and apply the same Lanne–Nyberg GFEVD formula as for the Burr model. Over $h = 0, \ldots, 10$, total connectedness is then 39.15%, compared with 40.04% for the Burr, while net EUR connectedness is +0.36 percentage points, compared with −1.32. Thus, the difference in total connectedness becomes substantially smaller when the response scale and share definition are aligned, whereas the net connectedness estimates remain sensitive to the model. The shocks remain model specific, and the comparison does not establish which model measures transmission more accurately.

[6]The benchmark model differs in the data transformation, as well as shock definition and decomposition, so this comparison is descriptive.

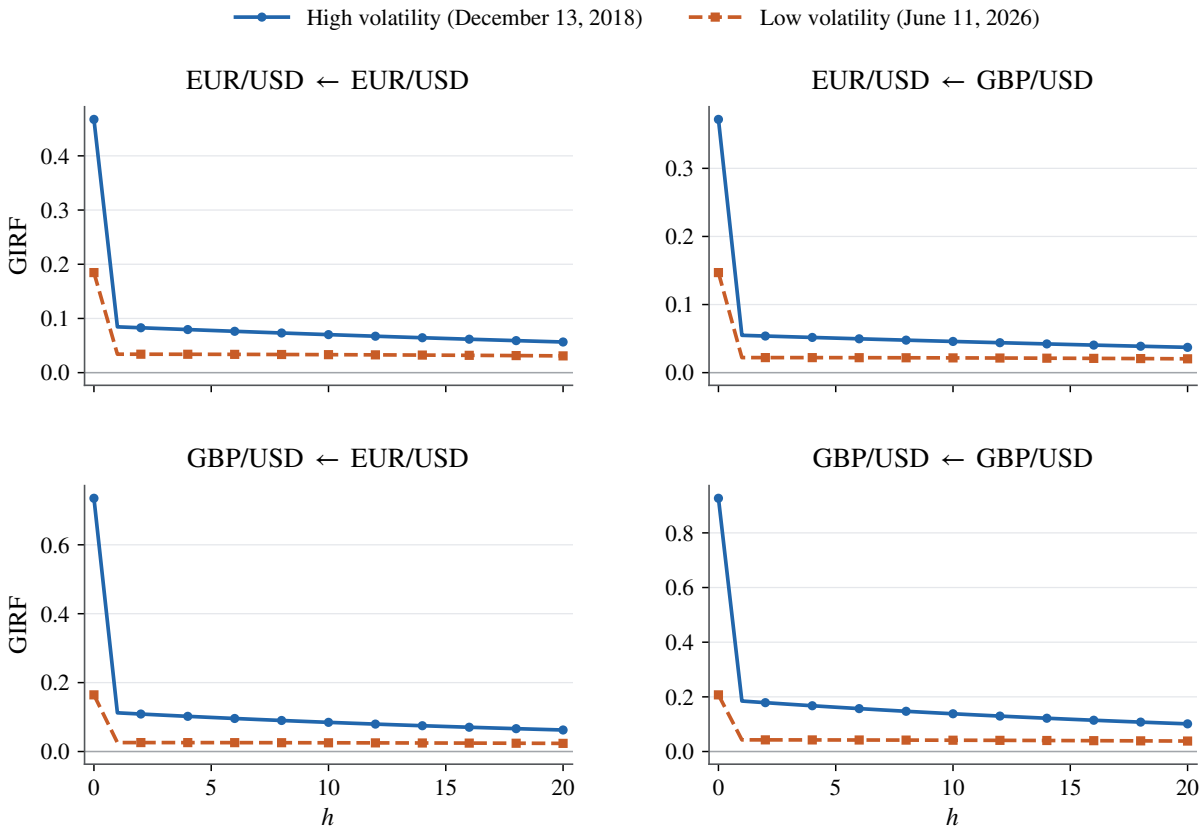


Figure 3: Generalized impulse responses to 90th-percentile shocks on a high-volatility day (December 13, 2018) and a low-volatility day (June 11, 2026), selected at the 90th and 10th percentiles of the average fitted log-scale deviation from the long-run level. Burr baseline specification ($\xi_1 = \xi_2 = 1$, diagonal Φ_λ and unrestricted K_λ), horizons $h = 0, \ldots, 20$.

Table 3: Connectedness between EUR/USD and GBP/USD

Horizon	EUR ← GBP	GBP ← EUR	Total	Net EUR (pp)
$h = 0$	44.24	43.74	43.99	−0.49
	(0.16)	(0.16)	(0.05)	(0.31)
$h = 0, \ldots, 10$	40.70	39.38	40.04	−1.32
	(0.13)	(0.13)	(0.03)	(0.25)
$h = 0, \ldots, 20$	39.96	38.50	39.23	−1.45
	(0.12)	(0.13)	(0.03)	(0.24)
$h = 1, \ldots, 10$	36.83	34.84	35.84	−1.99
	(0.16)	(0.16)	(0.04)	(0.31)
$h = 1, \ldots, 20$	36.90	34.91	35.91	−1.98
	(0.16)	(0.16)	(0.04)	(0.31)

Notes: The table reports sample averages of GFEVD shares and total connectedness (in %), computed under the Burr baseline specification ($\xi_1 = \xi_2 = 1$, diagonal Φ_λ and unrestricted K_λ) using $B = 32{,}000$ shock draws. Own-shock shares are omitted. For each target series, shares across all sources, including itself, sum to 100%. Net EUR is the difference between EUR→GBP and GBP→EUR, in percentage points. MC standard errors are in parentheses.

4.2 Application to futures markets

In the second part of the empirical application, we use 5-minute realized variances of four futures contracts from VOLARE, i.e., the S&P 500 E-mini (ES), crude oil (CL), natural gas (NG), and gold (GC).[7] The sample period is from April 28, 2010, to September 30, 2026, containing 4,242 daily observations. The time series means (standard deviations) are 2.78 (7.38) for ES, 18.59 (201.97) for CL, 25.31 (44.14) for NG, and 2.58 (4.16) for GC.[8] The pairwise correlations of the logarithms range from 0.26 (NG and GC) to 0.57 (ES and CL).

We estimate the model (1)–(4) under the multivariate Burr distribution, where $\xi_i = 1$, $\forall i$, in diagonal and baseline (diagonal Φ_λ and unrestricted K_λ) versions. The unrestricted K_λ raises the log-likelihood from $-38{,}209.25$ to $-37{,}821.05$, and BIC selects it. Table 4 reports the estimates under the baseline specification. The persistence parameters range from 0.94 for ES to 0.98 for NG, and the diagonal score coefficients from 0.06 for NG to 0.16 for ES. The fitted tail indices $\nu_{i\varsigma}$ are between 3.44 and 4.25, implying heavy upper tails.

Table 4: Estimation results under baseline specification

	ES	CL	NG	GC
$\log \nu_i$	0.86 (0.02)	1.08 (0.03)	0.97 (0.03)	0.93 (0.02)
$\omega_{\lambda,i}$	0.61 (0.09)	2.39 (0.11)	3.02 (0.09)	0.85 (0.08)
ϕ_{ii}	0.939 (0.006)	0.973 (0.005)	0.980 (0.004)	0.965 (0.005)
$k_{ES,\cdot}$	0.159 (0.011)	0.055 (0.007)	0.039 (0.006)	0.049 (0.008)
$k_{CL,\cdot}$	0.053 (0.006)	0.076 (0.008)	0.020 (0.004)	0.038 (0.006)
$k_{NG,\cdot}$	0.034 (0.005)	0.023 (0.005)	0.055 (0.006)	0.021 (0.004)
$k_{GC,\cdot}$	0.042 (0.007)	0.033 (0.006)	0.024 (0.006)	0.072 (0.009)
$\nu_{i\varsigma}$	3.44	4.25	3.82	3.66
$\log \varsigma$		0.371 (0.029)		
LLF		$-37{,}821.05$		
AIC		75,700.1		
BIC		75,884.3		

Notes: The table reports estimates of the model (1)–(4) under the Burr distribution ($\xi_i = 1, \forall i$) and baseline dynamic specification (diagonal Φ_λ and unrestricted K_λ) for ES, CL, NG, and GC realized variances. Robust Newey–West standard errors are in parentheses. The sample period is from April 28, 2010, to September 30, 2026, containing 4,242 daily observations.

Figure 4 reports the relative responses. A 90th-percentile shock raises a series' own realized variance by between 67% (CL) and 80% (ES) on impact. A similar shock to any other series raises the expected realized variance of ES by 38.6%, CL by 31.4%, NG by 34.9%, and GC by 36.4%. Beyond impact, the responses differ across source series. Responses to NG shocks are the smallest, between 4.7% and 9.1% at $h = 1$, and responses to ES shocks are the most persistent. In particular, the response of CL to ES is 11.9% at $h = 1$ and 7.1% at $h = 20$, compared with about 4.5% at $h = 20$ for responses to CL shocks. Figure 5 compares the generalized impulse responses on a high-volatility day (July 1, 2022) and a low-volatility day (May 6, 2019). The baseline forecasts on impact are between 1.7 (GC) and 11.4 (NG) times larger on the high-volatility day.

By Corollary 2, impact connectedness is constant over time with a total of 51.7%. Under the Burr specification, shocks at a common innovation percentile produce the same relative impact

[7] As in the exchange rate application, realized variances are multiplied by 252×100.

[8] VOLARE does not document how contract rolls are handled in the continuous futures series. On April 16, 2020, for example, the CL realized variance is about 33 times its bipower counterpart, suggesting that isolated large intraday returns dominate the measure. Without further information, we cannot determine whether these reflect market movements, contract rolls, or data problems. We do not adjust these observations.

response in a given target series across all other sources.[9] In this application, every estimated cross-series impact GFEVD share is close to 17%, and the net impact measures range from -0.62 to $+1.56$ percentage points. Over $h = 1, \ldots, 10$, Table 5 shows that ES is the largest net transmitter ($+29.3$ percentage points), followed by CL ($+9.8$), whereas NG (-26.4) and GC (-12.7) are net receivers. Total connectedness over this period is 45.4%. The cumulative measures over $h = 0, \ldots, 10$ show the same ordering with smaller magnitudes. Connectedness varies slightly over time, i.e., the cumulative total ranges from 47.8% to 50.4%, whereas the net measures over $h = 1, \ldots, 10$ vary by at most 0.54 percentage points.

We fit a VAR(5) model in log realized variances (Diebold and Yilmaz, 2012, 2014) as a benchmark. Applying the Lanne–Nyberg GFEVD formula to the generalized responses of realized variance in levels implied by the VAR under Gaussian innovations, over $h = 0, \ldots, 10$, gives net connectedness of $+14.4$ (ES), $+4.6$ (CL), -16.6 (NG), and -2.3 (GC) percentage points. The conventional generalized FEVD at the corresponding 11-day forecast horizon gives $+12.2$, $+0.7$, -9.1, and -3.9. In this application, the models identify the same net transmitters and receivers, although the magnitude of the net connectedness measures differs.

5 Limitations and extensions

The empirical applications illustrate the method in bivariate and $N = 4$ systems under the multivariate Burr distribution. The latter application demonstrates computational feasibility, but its empirical interpretation is relatively limited by the imposed restriction on the innovation distribution.

Estimating all shape parameters of the unrestricted MGB2 model proved difficult in our exploratory analyses, with some estimates approaching a limiting log-normal distribution. Related estimation difficulties for univariate GB2 score-driven models are discussed by Harvey and Palumbo (2023), who focus on Burr and symmetric GB2 specifications. We therefore use the more parsimonious Burr distribution in empirical applications. The shared gamma component links marginal shapes and contemporaneous dependence, which may limit empirical flexibility. The Burr restriction simplifies the estimation, but imposes additional restrictions on dependence and cross-series impact responses.

Extending the filter dynamics and developing uncertainty measures for connectedness are useful directions for further work. Time-varying shape parameters, discussed in Section 2.6, would also permit changes in contemporaneous dependence and relative responses.

6 Conclusion

We derive analytical generalized impulse responses for an MGB2 score-driven model for strictly positive time series and use them to construct connectedness measures. The generalized impulse response is the product of a baseline forecast, which captures the effect of market conditions, and a history-independent relative response, which describes the effect of the shock. With fixed shape parameters, connectedness on impact is constant over time. We also establish conditions under which cross-series responses vanish at particular horizons. The main practical implication is that connectedness measures should be interpreted alongside response magnitudes and horizons.

Simulations illustrate that including impact responses can obscure differences in transmission at later horizons. In the exchange rate application, connectedness is nearly constant over time, whereas responses to a shock at the same innovation quantile vary substantially across market conditions. In the futures application, we find that the S&P 500 E-mini and crude oil are net transmitters of volatility over the ten trading days after the shock, while natural gas and gold are

[9] This comes from the Burr distribution. With $\xi_i = 1$, $\forall i$, every innovation has the same quantile function in terms of $\varepsilon_{it}^{\nu_i}$, so the conditional expectations in (23) do not depend on the source series.

net receivers. Comparisons with a VAR benchmark show that differences between connectedness estimates can reflect the response scale and GFEVD construction, as well as the fitted dynamics.

Future work could allow shape parameters of the MGB2 distribution to vary over time, capturing changes in tail behavior, and assess how uncertainty in the estimated model parameters affects connectedness measures.

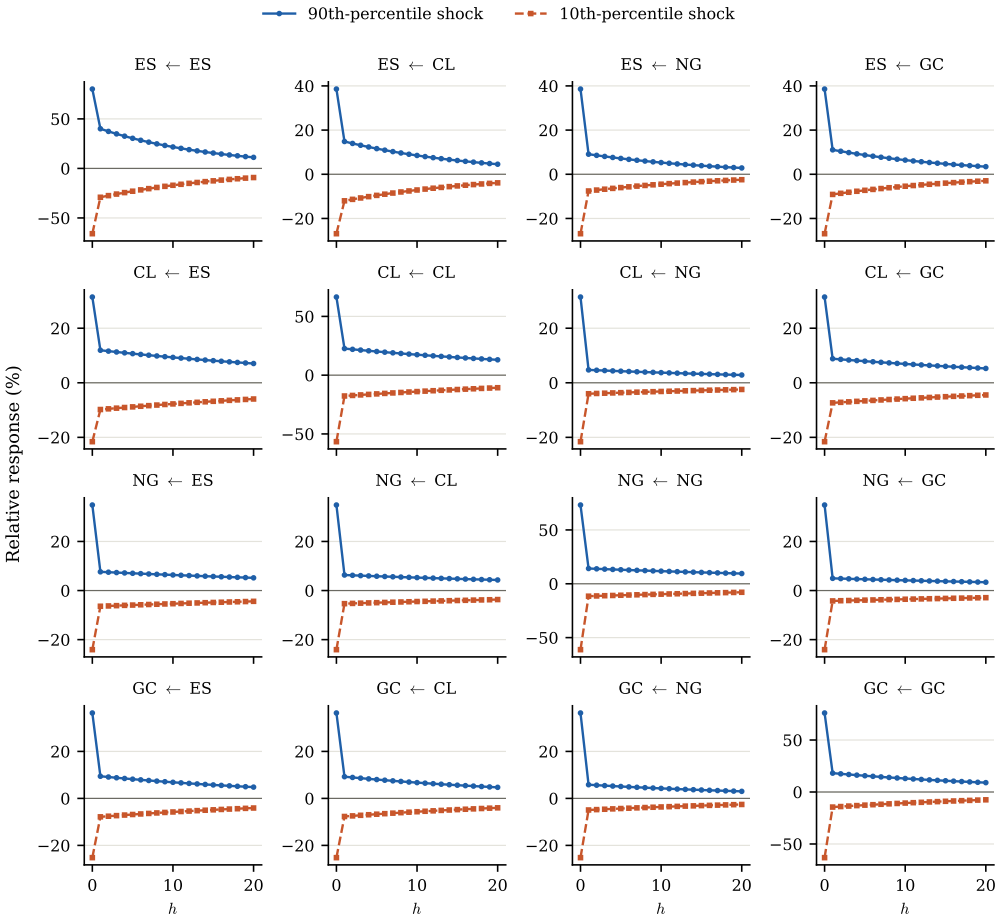


Figure 4: Relative responses $g_{ji}(h;\delta_i)$ to shocks at the 10th and 90th percentiles of each innovation, expressed as percentages of the baseline forecast. Burr baseline specification ($\xi_i = 1$, $\forall i$, diagonal Φ_λ and unrestricted K_λ), horizons $h = 0, \ldots, 20$. Rows are receivers and columns are source series.

Table 5: Connectedness between futures markets

Receiver \ Source	ES	CL	NG	GC	From others
ES	62.5	18.0	8.2	11.3	37.5
CL	24.1	55.3	5.3	15.2	44.7
NG	22.0	16.5	50.6	10.9	49.4
GC	20.7	20.0	9.4	49.9	50.1
To others	66.8	54.5	23.0	37.4	45.4
Net	+29.3	+9.8	−26.4	−12.7	
Net, $h = 0$	−0.6	+1.6	−0.4	−0.6	51.7
Net, $h = 0, \ldots, 10$	+13.4	+4.9	−12.4	−5.9	48.9
Net, log RV, LN shares in RV levels	+14.4	+4.6	−16.6	−2.3	44.7
Net, log RV, conventional (11 steps)	+12.2	+0.7	−9.1	−3.9	29.6

Notes: The table reports sample averages of GFEVD shares (in %), computed under the Burr baseline specification ($\xi_i = 1$, $\forall i$, diagonal Φ_λ and unrestricted K_λ) using $B = 32{,}000$ shock draws. "Net" is the difference between connectedness transmitted to the other series and connectedness received from them, in percentage points. The upper panel and the "To others" and "Net" rows report connectedness over $h = 1, \ldots, 10$, excluding impact responses. The final column reports connectedness received from other series in the upper panel and total connectedness in the "To others" row and the lower panel. The last two rows report results from a VAR(5) in log realized variances, using either generalized impulse responses of realized variance in levels and the Lanne–Nyberg GFEVD formula over $h = 0, \ldots, 10$, or the conventional generalized FEVD at an 11-day forecast horizon.

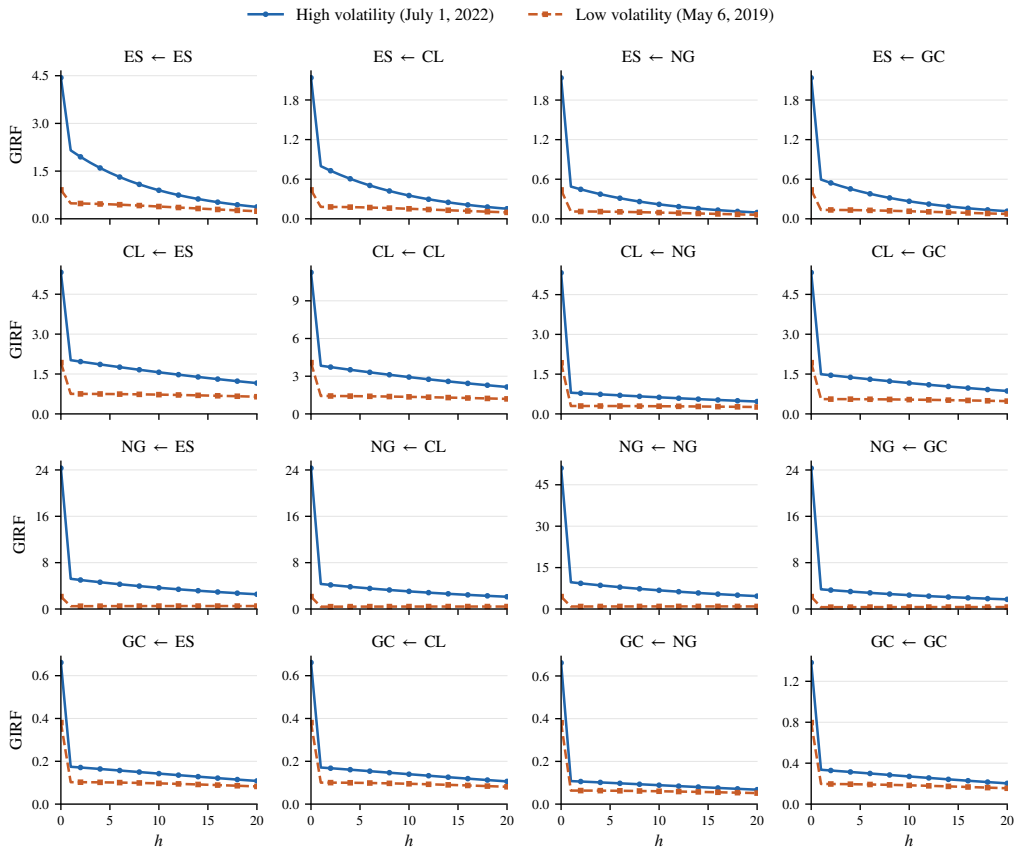


Figure 5: Generalized impulse responses to 90th-percentile shocks on a high-volatility day (July 1, 2022) and a low-volatility day (May 6, 2019), selected at the 90th and 10th percentiles of the average fitted log-scale deviation from the long-run level. Burr baseline specification ($\xi_i = 1$, $\forall i$, diagonal Φ_λ and unrestricted K_λ), horizons $h = 0, \ldots, 20$.

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A Derivation of the MGB2 forecasts

To derive the baseline MGB2 forecasts, we denote expectations conditional on the information available at time $t-1$ by $\mathbb{E}_{t-1}(\cdot) = \mathbb{E}(\cdot \mid \mathcal{F}_{t-1})$.

A.1 MGB2 moments and the Dirichlet moment generating function (MGF)

The MGB2 innovations admit the representation

$$\varepsilon_{it} = \left(\frac{G_{it}}{G_{0t}}\right)^{1/\nu_i}, \qquad G_{it} \sim \text{Gamma}(\xi_i, 1), \qquad G_{0t} \sim \text{Gamma}(\varsigma, 1), \tag{30}$$

where the gamma variables on the right-hand side are mutually independent $\forall t$. The denominator G_{0t} is common to all series. For $m = (m_1, \ldots, m_N)'$, independence of the gamma variables gives

$$C_m := \mathbb{E}\Big[\prod_{i=1}^{N} \varepsilon_{it}^{m_i}\Big] = \mathbb{E}\Big[G_{0t}^{-\sum_i m_i/\nu_i}\Big] \prod_{i=1}^{N} \mathbb{E}\Big[G_{it}^{m_i/\nu_i}\Big] = \frac{\Gamma\big(\varsigma - \sum_{i=1}^{N} m_i/\nu_i\big)}{\Gamma(\varsigma)} \prod_{i=1}^{N} \frac{\Gamma(\xi_i + m_i/\nu_i)}{\Gamma(\xi_i)}, \tag{31}$$

where $m_i > -\nu_i \xi_i$, $\forall i$, and $\sum_i m_i/\nu_i < \varsigma$.

Let us denote the selection vector with one as its jth element and zeros elsewhere by e_j. It follows from (31) that

$$\kappa_{q,j} := \mathbb{E}(\varepsilon_{jt}^{q}) = C_{qe_j} = \frac{\Gamma(\xi_j + q/\nu_j)\,\Gamma(\varsigma - q/\nu_j)}{\Gamma(\xi_j)\,\Gamma(\varsigma)}, \qquad \varsigma > q/\nu_j, \quad q > 0. \tag{32}$$

Thus, $\mu_j = \kappa_{1,j}$ in (6). The innovations have a positive mean, which is not necessarily equal to one.

To obtain the moment generating function (MGF) of the score, let us define $M_u(s) = \mathbb{E}[\exp(s'u_t)]$ for $s \in \mathbb{R}^N$. Under (3),

$$(b_{1t}, \ldots, b_{Nt}, b_{0t}) \sim \text{Dirichlet}(\xi_1, \ldots, \xi_N, \varsigma).$$

Let $|k| = \sum_{i=1}^{N} k_i$ for $k = (k_1, \ldots, k_N) \in \mathbb{N}_0^N$, and let $(a)_q = \Gamma(a+q)/\Gamma(a)$. Dirichlet moments satisfy

$$\mathbb{E}\Big[\prod_{i=1}^{N} b_{it}^{k_i}\Big] = \frac{\prod_{i=1}^{N} (\xi_i)_{k_i}}{(c)_{|k|}}, \qquad c = \varsigma + \sum_{i=1}^{N} \xi_i. \tag{33}$$

Expanding the exponential and using the score representation (4) therefore yields

$$M_u(s) = \exp(-s' D_\nu \xi)\, \mathbb{E}\Big[\exp\Big(c \sum_{i=1}^{N} \nu_i s_i b_{it}\Big)\Big] = \exp(-s' D_\nu \xi) \sum_{k \in \mathbb{N}_0^N} \frac{\prod_{i=1}^{N} (\xi_i)_{k_i}}{(c)_{|k|}} \prod_{i=1}^{N} \frac{(c\nu_i s_i)^{k_i}}{k_i!}. \tag{34}$$

A.2 Multi-step-ahead forecasts

Iterating forward (2) implies

$$\lambda_{t+h|t+h-1} = \omega_\lambda + \Phi_\lambda^h(\lambda_{t|t-1} - \omega_\lambda) + \sum_{r=0}^{h-1} \Phi_\lambda^r K_\lambda u_{t+h-1-r}. \tag{35}$$

Taking expectations,

$$\bar{\lambda}_t(h) := \mathbb{E}_{t-1}(\lambda_{t+h|t+h-1}) = \omega_\lambda + \Phi_\lambda^h(\lambda_{t|t-1} - \omega_\lambda), \tag{36}$$

so that $a_{j,t}(h) = e_j'\bar{\lambda}_t(h)$ in (15). For $h = 0$, $\bar{\lambda}_t(0) = \lambda_{t|t-1}$, which is $\mathcal{F}_{t-1}$-measurable. Using $s_j(r)$ from (15), let us apply the MGF in (34) to future scores in (35). As a result,

$$\mathbb{E}_{t-1}(\alpha_{j,t+h|t+h-1}) = \exp\{e_j'\bar{\lambda}_t(h)\}\prod_{r=0}^{h-1} M_u(s_j(r)). \tag{37}$$

Taken together, equations (1), (31), and (35) imply

$$\mathbb{E}_{t-1}\Big[\prod_{i=1}^{N} y_{i,t+h}^{m_i}\Big] = C_m \exp\{m'\bar{\lambda}_t(h)\}\prod_{r=0}^{h-1} M_u\big(K_\lambda'(\Phi_\lambda')^r m\big), \tag{38}$$

provided the moment C_m exists. Taking $m_j = 1$ and $m_i = 0$, $\forall i \neq j$, gives the mean forecast

$$\mathbb{E}_{t-1}(y_{j,t+h}) = \kappa_{1,j}\exp\{e_j'\bar{\lambda}_t(h)\}\prod_{r=0}^{h-1} M_u(s_j(r)), \qquad \varsigma > 1/\nu_j. \tag{39}$$

Correspondingly, taking $m_j = 2$ and $m_i = 0$, $\forall i \neq j$, gives the second moment forecast

$$\mathbb{E}_{t-1}(y_{j,t+h}^2) = \kappa_{2,j}\exp\{2e_j'\bar{\lambda}_t(h)\}\prod_{r=0}^{h-1} M_u(2s_j(r)). \tag{40}$$

Therefore, the variance forecast is given by

$$\mathrm{Var}_{t-1}(y_{j,t+h}) = \exp\{2e_j'\bar{\lambda}_t(h)\}\left[\kappa_{2,j}\prod_{r=0}^{h-1} M_u(2s_j(r)) - \kappa_{1,j}^2\Big\{\prod_{r=0}^{h-1} M_u(s_j(r))\Big\}^2\right], \qquad \varsigma > 2/\nu_j. \tag{41}$$

A.3 Diagonal parameterization

If $\Phi_\lambda = \mathrm{diag}(\phi_1, \ldots, \phi_N)$ and $K_\lambda = \mathrm{diag}(k_1, \ldots, k_N)$, then $s_j(r) = k_j\phi_j^r e_j$. Only the marginal distribution of b_{jt} is needed, i.e.,

$$b_{jt} \sim \mathrm{Beta}(\xi_j, c - \xi_j), \qquad u_{jt} = \nu_j(c\, b_{jt} - \xi_j).$$

Hence, the MGF required for the forecasts becomes

$$M_{u_j}(z) = \exp(-z\nu_j\xi_j)\sum_{q=0}^{\infty}\frac{(\xi_j)_q}{(c)_q}\frac{(c\nu_j z)^q}{q!}. \tag{42}$$

B Proofs for Section 2.4

Proof of Lemma 1. Let us consider the representation (30) and define $T_{it} = G_{it} + G_{0t} \sim \mathrm{Gamma}(\xi_i + \varsigma, 1)$ and $q_{it} = G_{it}/T_{it} \sim \mathrm{Beta}(\xi_i, \varsigma)$. By Lukacs (1955), q_{it} and T_{it} are independent. Since both are functions of (G_{it}, G_{0t}) only, the pair (q_{it}, T_{it}) is also independent of $G_{-i,t} = (G_{jt})_{j\neq i}$. Hence q_{it}, T_{it} and $G_{-i,t}$ are mutually independent. For $j \neq i$, we have $b_{jt} = G_{jt}/(T_{it} + \sum_{l\neq i} G_{lt})$, so the vector $b_{-i,t}$ is a function of $(T_{it}, G_{-i,t})$ only and is therefore independent of q_{it}. As $\varepsilon_{it} = \{q_{it}/(1-q_{it})\}^{1/\nu_i}$ is a strictly increasing function of q_{it}, $b_{-i,t}$ is independent of ε_{it} as well. Finally, by (4), $u_{-i,t}$ is a function of $b_{-i,t}$ only, so it is also independent of ε_{it}.

Proof of (19). From (30), $G_{0t} = (1-q_{it})T_{it} = T_{it}/(1+\varepsilon_{it}^{\nu_i})$, so that, for $j \neq i$, $\varepsilon_{jt} = (1+\varepsilon_{it}^{\nu_i})^{1/\nu_j}(G_{jt}/T_{it})^{1/\nu_j}$. Since G_{jt}, T_{it} and q_{it} are mutually independent,

$$\mathbb{E}(\varepsilon_{jt} \mid \varepsilon_{it} = x) = (1+x^{\nu_i})^{1/\nu_j}\,\mathbb{E}\big(G_{jt}^{1/\nu_j}\big)\,\mathbb{E}\big(T_{it}^{-1/\nu_j}\big),$$

and the gamma moments give (19), provided $\xi_i + \varsigma > 1/\nu_j$.

Proof of Proposition 1. At $h = 0$, $y_{jt} = \alpha_{j,t|t-1}\varepsilon_{jt}$ with $\alpha_{j,t|t-1}$ $\mathcal{F}_{t-1}$-measurable and $m_{j,t}(0) = \mu_j \alpha_{j,t|t-1}$, which gives (22)–(23). For $h \geq 1$, (35) gives

$$e_j' \lambda_{t+h|t+h-1} = a_{j,t}(h) + \sum_{r=0}^{h-1} s_j(r)' u_{t+h-1-r},$$

so u_t enters only through the term $r = h-1$. The future score vectors $u_{t+1}, \ldots, u_{t+h-1}$ are i.i.d. and independent of ε_{it}. The innovation ε_{t+h} is independent of all preceding scores and of ε_{it}. Hence,

$$\mathbb{E}\big[y_{j,t+h} \mid \mathcal{F}_{t-1}, z_{it} = \delta_i\big] = \mu_j \exp(a_{j,t}(h))\, M_u^{(i)}\big(s_j(h-1); q_i\big) \prod_{r=0}^{h-2} M_u\big(s_j(r)\big),$$

where $M_u^{(i)}$ follows from (4) and (10)–(11). Dividing by (17) gives (24).

Proof of Corollary 1. The ith component of $s_j(h-1) = K_\lambda'(\Phi_\lambda')^{h-1} e_j$ is $(\Phi_\lambda^{h-1} K_\lambda)_{ji}$. If it is zero, then $s_j(h-1)' u_t$ depends only on $u_{-i,t}$. By Lemma 1, the vector $u_{-i,t}$ is independent of ε_{it} and hence of q_{it}. Therefore,

$$M_u^{(i)}\big(s_j(h-1); q_i\big) = M_u\big(s_j(h-1)\big),$$

so $g_{ji}(h; \delta_i) = 0$ given (24). The response is then zero by (21).

Proof of Corollary 3. With diagonal Φ_λ and K_λ, $s_j(r) = k_j \phi_j^r e_j$, so $s_j(r)' u_t = k_j \phi_j^r u_{jt}$. For $j \neq i$, Lemma 1 implies that the distribution of u_{jt} does not depend on q_{it}, so $M_u^{(i)}(s_j(r); q) = M_u(s_j(r))$ and $g_{ji}(h; \delta_i) = 0$ for $h \geq 1$.

C Monte Carlo assessment of the analytical responses

This appendix evaluates the numerical implementation of the analytical generalized impulse responses in Proposition 1. We compare them with independently simulated responses, check the conditional innovation sampler in (7)–(12), and assess the MC precision of the generalized shares. We also verify the properties implied by Lemma 1 and Corollaries 2–3.

C.1 Simulation design

We consider $N = 3$ series with shape parameters $\xi = (1.5, 2.0, 1.2)'$ and $\nu = (1.8, 1.3, 2.2)'$, and set $\omega_\lambda = (0.2, -0.1, 0.3)'$, under three specifications. In the first, the score-driven recursion has diagonal parameter matrices, with $\Phi_\lambda = \mathrm{diag}(0.95, 0.90, 0.85)$ and $K_\lambda = \mathrm{diag}(0.05, 0.04, 0.06)$, while the parameter $\varsigma = 3$. In the second, we allow for non-diagonal dynamics, with $\varsigma = 3$ and

$$\Phi_\lambda = \begin{pmatrix} 0.93 & 0.03 & 0 \\ 0.02 & 0.90 & 0.03 \\ 0 & 0.04 & 0.88 \end{pmatrix}, \qquad K_\lambda = \begin{pmatrix} 0.05 & 0.02 & 0.01 \\ 0.02 & 0.04 & 0.02 \\ 0.03 & 0.01 & 0.06 \end{pmatrix}.$$

The third specification features the same non-diagonal dynamics, but we set $\varsigma = 2$, implying heavier tails. All three specifications satisfy $\varsigma > 2/\nu_j$ $\forall j$, ensuring finite second moments.

For each source series i, we consider positive and negative shocks of the form $\delta_i = \varepsilon_i^{(p)} - \mu_i$, where $\varepsilon_i^{(p)}$ denotes the pth quantile of the marginal innovation distribution. We use $p \in \{0.10, 0.90\}$ in the first two specifications and $p \in \{0.01, 0.99\}$ in the third. The responses are evaluated at horizons $h = 0, 1, \ldots, 10$ for two initial log-scale vectors $\lambda_{t|t-1} = \omega_\lambda$ and $\lambda_{t|t-1} = \omega_\lambda + (0.5, -0.4, 0.3)'$.

We calculate the analytical responses using (21)–(24). The MGFs in (18) and (20) are numerically evaluated using the Gauss–Beta quadrature with $Q = 32$ nodes for each Beta integral arising from the Dirichlet representation.

The simulation does not use Proposition 1. To evaluate the first expectation in (13), we generate ε_t conditional on $\varepsilon_{it} = \mu_i + \delta_i$ using two different samplers. The first is given by (7)–(12). The second is obtained directly from the gamma representation in (30). The joint density of (G_{0t}, G_{it}) is proportional to $g_0^{\varsigma-1} g_i^{\xi_i-1} \exp(-g_0 - g_i)$. A change of variables and conditioning on $G_{it}/G_{0t} = x^{\nu_i}$ give

$$G_{0t} \mid \{\varepsilon_{it} = x\} \sim \text{Gamma}\big(\xi_i + \varsigma,\ 1 + x^{\nu_i}\big),$$

where the parameters denote shape and rate, respectively. Conditional on $\varepsilon_{it} = x$, this variable is independent of the remaining G_{jt}, $j \neq i$, which retain their original gamma distributions. Setting $G_{it} = x^{\nu_i} G_{0t}$ then gives the conditional innovation vector. This construction provides a separate check on the sampler in (7)–(12), without relying on Dirichlet aggregation.

To evaluate the second expectation in (13), we draw ε_t from the unconditional gamma representation in (30). We generate the shocked and baseline paths recursively using (2), calculating the scores from the simulated observations according to (3)–(4). Both paths use the same future innovation vectors at $t+1, \ldots, t+h$. Each response is estimated from $R = 2 \times 10^5$ replications, with a separate random seed for each combination of initial log-scale vector, shock source, shock sign, and sampling method.

C.2 Results

Table 6 summarizes the results. The differences between analytical and simulated responses are consistent with their MC standard errors across both samplers, both initial log-scale vectors, and all three specifications.[10] Figure 6 illustrates this comparison for shocks to the first series under non-diagonal dynamics, showing the differences relative to pointwise 95% MC bands. The simulated baseline forecasts are also close to the analytical forecasts in (17).

The standardized differences, denoted by z, are the differences divided by their estimated MC standard errors. They are correlated across horizons because observations along each simulated path share the same initial innovation draw ε_t. The comparisons therefore cannot be treated as independent tests, and the proportion satisfying $|z| \leq 1.96$ does not necessarily have to be exactly 95%. To check whether the differences result from MC variation, we repeat the three cases with the largest $|z|$ in each specification using $R = 10^6$ replications and two new seeds. A fixed non-zero bias would tend to become more apparent as the MC standard error decreases. Instead, the largest $|z|$ is smaller in the reruns for each specification.

In the third specification, $\varsigma < 4/\nu_2$, so the fourth moment of y_{2t} does not exist. Although second moments are finite, estimated MC standard errors can be unstable, and the normal approximation may be less reliable at the sample sizes considered. We therefore examine more closely the case in which the response differences most consistently have the same sign across horizons, i.e., a positive shock to series 3 at its 99th-percentile value, starting from $\lambda_{t|t-1} = \omega_\lambda$. Across runs with 10^6, 4×10^6 and 1.6×10^7 replications, the largest $|z|$ ranges from 2.0 to 2.8, while the standardized differences averaged across horizons change sign. In the largest run, the maximum absolute response difference is approximately 8×10^{-4}. Taken together, these results support MC variation as an explanation for the observed differences.

We then compare the conditional innovation distributions directly, using 10^6 draws from each conditional sampler and an additional comparison based on rejection sampling. For the latter, we generate 5×10^7 unconditional innovation vectors from (30) and keep those satisfying $|\log \varepsilon_{it} - \log x| < 0.005$. This check does not use either conditional sampling formula. It approximates conditioning at x by conditioning on a small window around it, so the differences may reflect both MC variation and the finite window width.

[10] Doubling the number of quadrature nodes from 32 to 64 changes the analytical responses by at most 2×10^{-15}.

Table 6: MC assessment of the analytical responses and GFEVD shares

	Diagonal	Non-diagonal ($\varsigma = 3$)	Non-diagonal ($\varsigma = 2$)
Analytical vs. simulated GIRFs			
max $\lvert\text{GIRF}(Q) - \text{GIRF}(2Q)\rvert$	1.7×10^{-15}	1.8×10^{-15}	2.0×10^{-15}
max $\lvert\text{analytical} - \text{simulated}\rvert$	5.3×10^{-3}	6.7×10^{-3}	3.0×10^{-2}
max MC s.e.	3.3×10^{-3}	3.3×10^{-3}	1.1×10^{-2}
max $\lvert z\rvert$, sampler (7)–(12)	2.75	2.85	3.14
max $\lvert z\rvert$, gamma-conditional sampler	3.21	3.41	3.11
share with $\lvert z\rvert \leq 1.96$	0.94	0.92	0.94
max $\lvert z\rvert$, reruns with $R = 10^6$	2.50	2.33	2.93
baseline forecast (17): max $\lvert z\rvert$	3.08	3.13	3.06
Conditional distribution given ε_{it}			
vs. gamma-conditional: max $\lvert z\rvert$	1.59	1.66	3.24
vs. gamma-conditional: KS rejections at 5%	2 of 12	0 of 12	2 of 12
vs. rejection sampling: max $\lvert z\rvert$	2.42	3.08	4.02
rejection sampling: min accepted draws	99,992	100,012	11,770
cases with $\lvert z\rvert > 3$; max $\lvert z\rvert$ on rerun	none	1; 2.59	2; 3.00
Checks of theoretical properties			
zero-mean identity: max $\lvert\mathbb{E}_\delta\, g\rvert$	1.3×10^{-15}	1.3×10^{-15}	1.8×10^{-14}
max $\lvert\text{row sum} - 1\rvert$	1.1×10^{-16}	1.1×10^{-16}	1.1×10^{-16}
$H = 0$: max difference in shares between histories	1.1×10^{-16}	2.8×10^{-17}	0
$H = 10$: max difference in shares between histories	4.3×10^{-3}	7.4×10^{-3}	4.3×10^{-3}
cross multipliers, $j \neq i$, $h \geq 1$: max $\lvert g_{ji}\rvert$	1.1×10^{-15}	0.164	0.106
Computing time (sec.)			
analytical	0.65	0.47	0.50
simulation, $R = 2 \times 10^5$	4.0	3.3	3.7

Notes: $N = 3$, $H = 10$, $Q = 32$ quadrature nodes for each Beta integral. GIRF maxima are taken over two initial log-scale vectors (histories), sources, targets, shock signs, horizons and, unless stated, both conditional samplers. z is the difference divided by its MC standard error. Reruns use the three cases with the largest $\lvert z\rvert$ in each specification, each with two new seeds. The conditional distribution statistics compare the sampler (7)–(12) with the alternatives indicated, using the means of $\log \varepsilon_{jt}$ and b_{jt}, $j \neq i$, for every source and shock. The two-sample KS tests compare draws of ε_{jt} from the two conditional samplers. GFEVD shares are based on $B = 2{,}000$ shock draws, common to both histories. Computing times refer to a Python implementation, cover all six GIRFs at one history, and exclude the GFEVD.

We compare the means of $\log \varepsilon_{jt}$ and b_{jt} for $j \neq i$, whose moments are finite, and apply two-sample Kolmogorov–Smirnov tests to the draws of ε_{jt} from the two conditional samplers. The tests reject in two of the twelve comparisons in the first and third specifications, and in none in the second. Isolated rejections are possible even when the conditional distributions are identical, given the number of tests performed.

In the third specification, the smallest accepted sample contains approximately 12,000 vectors out of the 5×10^7 unconditional draws. For the shock corresponding to the 99th percentile of ε_{2t} in the third specification, we repeat the rejection sampling comparison ten times using independent draws. For each of the four statistics, we calculate the mean and standard deviation of its z-scores across the ten runs. The means range from -0.14 to 0.24, and the standard deviations range from 1.08 to 1.22. The z-scores are close to zero on average, while their sample standard deviations are slightly above one.

The numerical checks confirm the theoretical properties. In particular, averaging the response multipliers over the distribution of shock sizes gives zero. This follows from (13), i.e., averaging the first conditional expectation over the distribution of $x = \varepsilon_{it}$ recovers the baseline expectation by the law of iterated expectations. Their difference therefore averages to zero. Since the baseline forecast $m_{j,t}(h)$ does not depend on the size of the shock, the multiplier g_{ji} also averages to zero. This is a useful consistency check, although it cannot establish correctness on its own because identically zero responses would also satisfy it. The generalized forecast error

variance shares sum to one across sources. Using the same shock draws for both initial log-scale vectors, we obtain identical shares at $H = 0$, as implied by Corollary 2, but different shares at $H = 10$. With diagonal coefficient matrices, a shock does not affect the other series at horizons $h \geq 1$, as implied by Corollary 3. Such effects are present in the non-diagonal specifications.

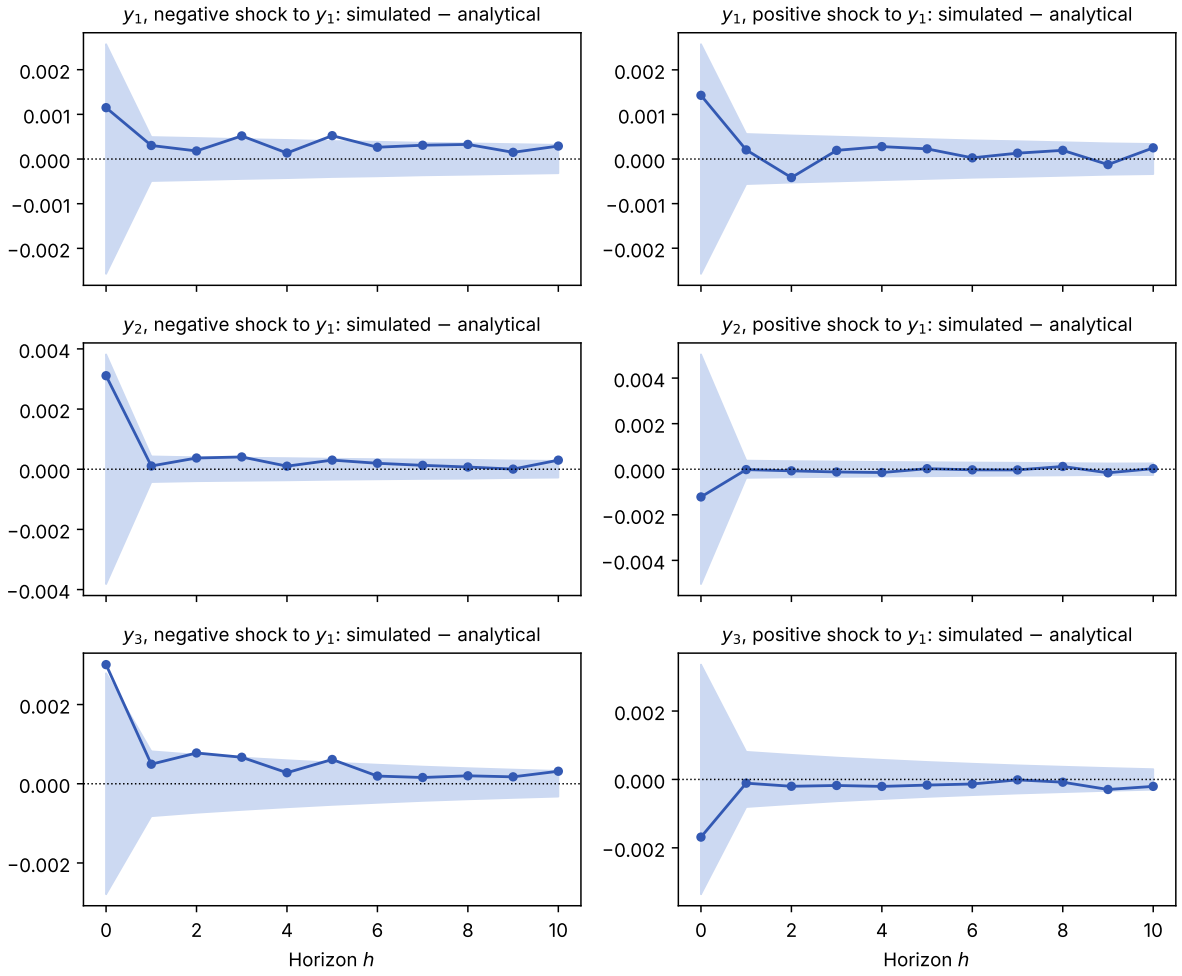


Figure 6: Difference between simulated and analytical generalized impulse responses to a negative (left) and positive (right) shock to series $i = 1$, non-diagonal dynamics with $\varsigma = 3$, $\lambda_{t|t-1} = \omega_\lambda$. Shaded areas are pointwise 95% MC bands.

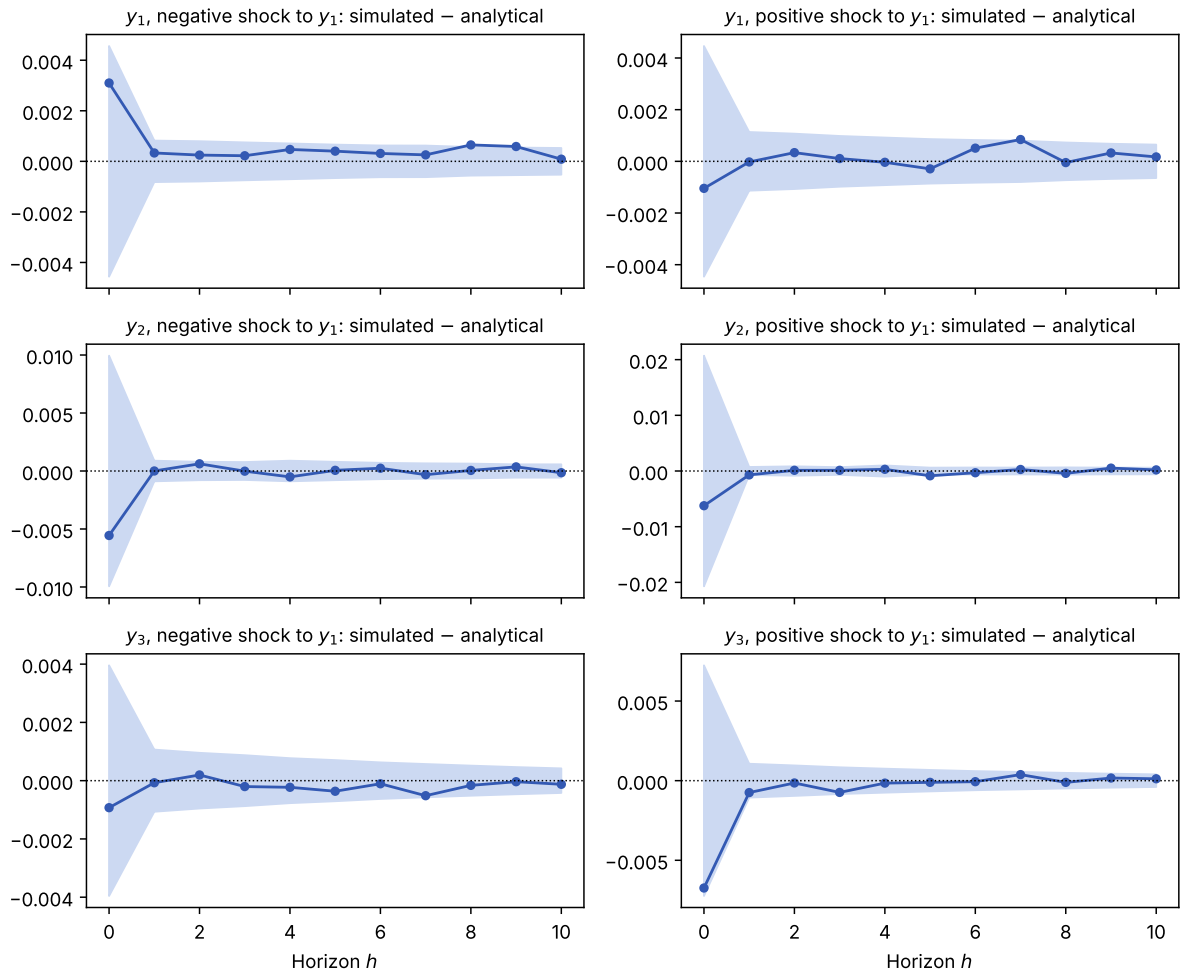


Figure 7: As in Figure 6, but for the non-diagonal specification with $\varsigma = 2$.

C.3 MC precision of the generalized shares

The GFEVD shares in (27) are calculated from the analytical GIRFs in Proposition 1 and averaged over B draws of the shock vector. We estimate the MC standard error of each share by

$$\widehat{\mathrm{se}}\big(\hat{\theta}^{(H)}_{ji,t}\big) = \frac{\mathrm{sd}_b\big(\lambda^{(H)}_{ji,t}(\delta^{(b)})\big)}{\sqrt{B}}.$$

Table 7 reports the largest standard error across target series, source series, and horizons. It decreases by approximately half whenever B increases four times, with similar values across the three specifications. At $B = 32{,}000$, the largest standard error is approximately 0.0018, or 0.18 percentage points. The corresponding pointwise 95% MC interval has a half-width of approximately 0.35 percentage points. We use the same shock draws for both histories, which pairs the share estimates and can reduce the MC standard error of their difference.

Table 7: MC standard errors of the GFEVD shares

	Diagonal		Non-diagonal ($\varsigma = 3$)		Non-diagonal ($\varsigma = 2$)	
B	max s.e.	max diff.	max s.e.	max diff.	max s.e.	max diff.
500	0.0143	0.0088	0.0144	0.0136	0.0143	0.0337
2,000	0.0072	0.0052	0.0071	0.0182	0.0071	0.0124
8,000	0.0036	0.0054	0.0036	0.0076	0.0035	0.0044
32,000	0.0018	–	0.0018	–	0.0018	–

Notes: Maxima over all target series j, source series i, and horizons $H \in \{0, 1, 5, 10\}$ at $\lambda_{t|t-1} = \omega_\lambda$. "max diff." is $\max |\hat{\theta}_B - \hat{\theta}_{32,000}|$. Draws are nested, so $\hat{\theta}_B$ uses the first B of the 32,000 draws.

C.4 Computational cost

For a given parameter vector and horizon, we first compute the quadrature nodes and weights used to calculate the score MGFs. These are reused across filtered histories and shock sizes, making subsequent evaluation of the response multipliers relatively inexpensive. In this experiment, the initial setup and evaluation of the MGFs and all six source–shock combinations at a given history take approximately half a second. Simulating the same six combinations with $R = 2 \times 10^5$ replications and the Dirichlet conditional sampler takes 3–4 seconds. The maximum standard errors are approximately 3×10^{-3} in the first two specifications and 1.1×10^{-2} in the third.

The timing and numerical accuracy results reported here are based on $N = 3$ series. Section 4.2 demonstrates computational feasibility for four series, but performance in substantially larger systems remains to be investigated. With tensor-product quadrature, evaluating M_u requires about Q^N node combinations, while each conditional kernel requires about Q^{N-1}. Thus, the computational cost grows exponentially with N. When some elements of $s_j(r)$ are zero, the Dirichlet aggregation property can reduce the integration dimension by combining the corresponding components. Sparse grid and quasi-MC methods may offer further alternatives to tensor-product quadrature. We leave these extensions for future work.

All computations were performed in Python, and the replication code is available upon request.

D Additional results

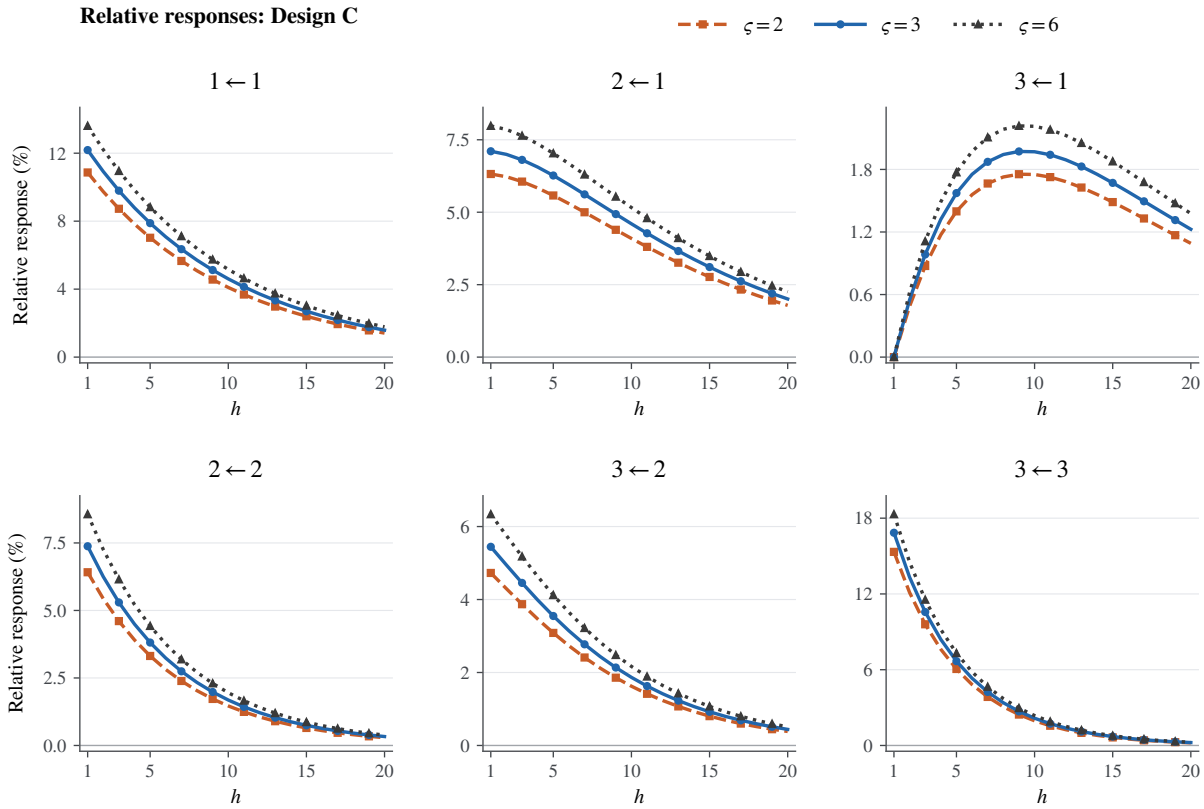


Figure 8: Relative responses $g_{ji}(h;\delta_i)$ at horizons $h = 1,\ldots,20$ to 90th-percentile shocks in Design C with $\varsigma = 2,3,6$, expressed as percentages of the baseline forecast.

Table 8: Connectedness in the controlled settings

Design	Horizon	$2 \leftarrow 1$	$3 \leftarrow 2$	$3 \leftarrow 1$	$1 \leftarrow 2$	$2 \leftarrow 3$	$1 \leftarrow 3$	Total
A	$h = 0$	20.7	19.1	16.5	21.2	18.8	16.6	37.6
	$h \leq 1$	20.6	18.7	16.1	21.0	18.7	16.4	37.2
	$0 \leq h \leq 5$	20.5	18.2	15.6	20.5	18.5	16.0	36.4
	$0 \leq h \leq 10$	20.4	18.1	15.5	20.2	18.5	15.8	36.2
	$0 \leq h \leq 20$	20.4	18.1	15.5	20.1	18.5	15.7	36.1
	$h = 1$	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	$1 \leq h \leq 5$	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	$1 \leq h \leq 10$	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	$1 \leq h \leq 20$	0.0	0.0	0.0	0.0	0.0	0.0	0.0
B	$h = 0$	20.7	19.1	16.5	21.2	18.8	16.6	37.6
	$h \leq 1$	21.0	19.3	15.9	21.0	18.5	16.4	37.4
	$0 \leq h \leq 5$	21.6	19.5	15.2	20.5	18.2	16.0	37.0
	$0 \leq h \leq 10$	21.8	19.5	15.1	20.2	18.1	15.8	36.8
	$0 \leq h \leq 20$	21.8	19.5	15.1	20.1	18.0	15.7	36.8
	$h = 1$	48.0	23.5	0.0	0.0	0.0	0.0	23.9
	$1 \leq h \leq 5$	48.0	23.6	0.0	0.0	0.0	0.0	23.9
	$1 \leq h \leq 10$	48.0	23.6	0.0	0.0	0.0	0.0	23.9
	$1 \leq h \leq 20$	48.0	23.6	0.0	0.0	0.0	0.0	23.9
C	$h = 0$	20.7	19.1	16.5	21.2	18.8	16.6	37.6
	$h \leq 1$	21.0	19.3	15.9	21.0	18.5	16.4	37.4
	$0 \leq h \leq 5$	22.2	19.9	15.2	20.5	18.0	16.0	37.3
	$0 \leq h \leq 10$	23.1	20.2	15.3	20.2	17.6	15.8	37.4
	$0 \leq h \leq 20$	23.8	20.2	15.7	20.1	17.4	15.7	37.6
	$h = 1$	48.0	23.5	0.0	0.0	0.0	0.0	23.9
	$1 \leq h \leq 5$	54.2	25.9	2.7	0.0	0.0	0.0	27.6
	$1 \leq h \leq 10$	58.4	26.1	6.8	0.0	0.0	0.0	30.4
	$1 \leq h \leq 20$	61.1	25.1	11.1	0.0	0.0	0.0	32.4
C ($\varsigma = 2$)	$h = 0$	24.7	23.0	20.5	24.9	23.0	20.6	45.6
	$h \leq 1$	24.9	22.9	20.0	24.7	22.8	20.4	45.3
	$0 \leq h \leq 5$	25.5	23.1	19.4	24.2	22.4	20.0	44.9
	$0 \leq h \leq 10$	26.1	23.2	19.4	24.0	22.2	19.8	44.9
	$0 \leq h \leq 20$	26.5	23.2	19.6	23.9	22.0	19.7	44.9
	$h = 1$	48.6	22.4	0.0	0.0	0.0	0.0	23.7
	$1 \leq h \leq 5$	55.0	24.8	2.5	0.0	0.0	0.0	27.5
	$1 \leq h \leq 10$	59.3	25.0	6.6	0.0	0.0	0.0	30.3
	$1 \leq h \leq 20$	62.0	24.1	10.8	0.0	0.0	0.0	32.3
C ($\varsigma = 6$)	$h = 0$	13.8	12.7	10.4	14.4	12.2	10.5	24.7
	$h \leq 1$	14.9	13.6	9.5	14.2	11.8	10.3	24.7
	$0 \leq h \leq 5$	18.0	15.7	8.5	13.7	10.9	10.0	25.6
	$0 \leq h \leq 10$	20.2	16.4	9.1	13.5	10.3	9.8	26.4
	$0 \leq h \leq 20$	21.5	16.4	10.1	13.4	10.0	9.7	27.0
	$h = 1$	47.3	24.9	0.0	0.0	0.0	0.0	24.1
	$1 \leq h \leq 5$	53.4	27.3	2.8	0.0	0.0	0.0	27.8
	$1 \leq h \leq 10$	57.4	27.4	7.1	0.0	0.0	0.0	30.6
	$1 \leq h \leq 20$	60.0	26.4	11.4	0.0	0.0	0.0	32.6

Notes: Entries are expressed in percent and evaluated at history $\lambda^{(a)}$, using $B = 32{,}000$ shock draws. The six directional columns report GFEVD shares; Total reports total connectedness. Within each design, the first five rows include impact responses, while the final four exclude them and use a separate normalization. The row $h = 0$ reports impact connectedness, while $h = 1$ reports only the responses at horizon 1. For each target series, the shares across all sources sum to 100%. The shares associated with its own shocks are omitted from the table.

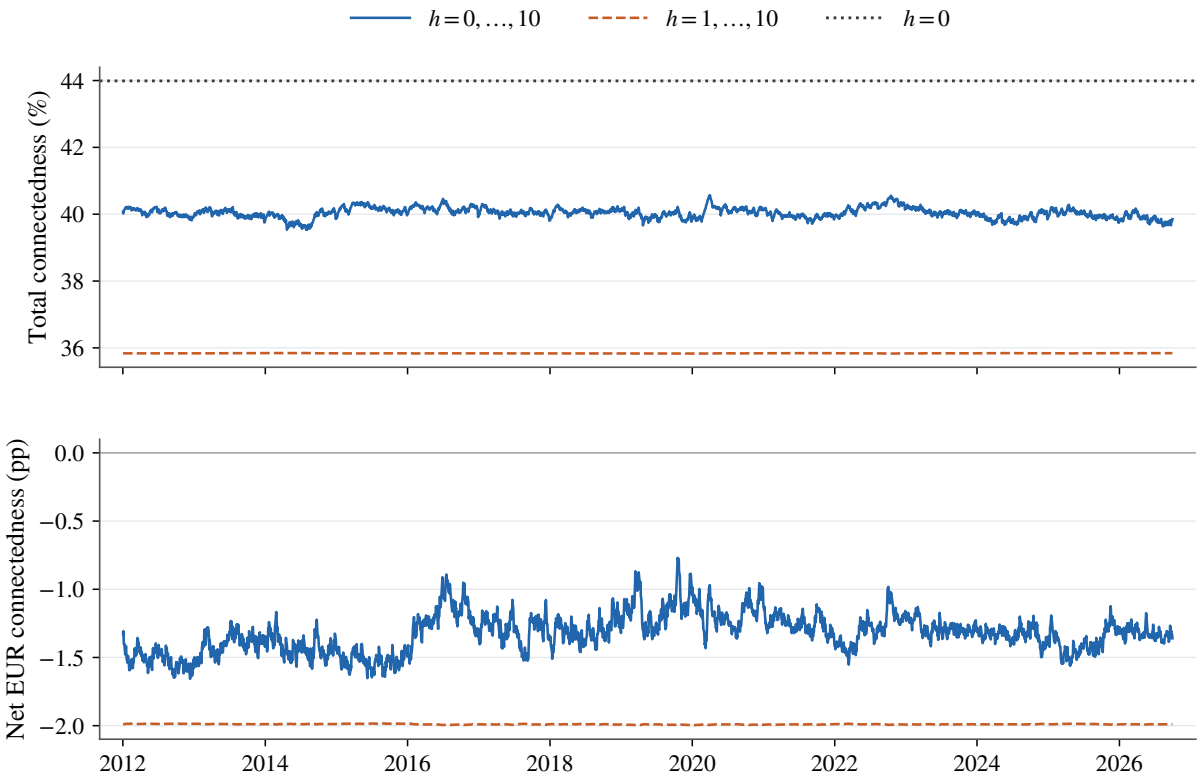


Figure 9: Total connectedness and net connectedness of EUR/USD over time, Burr baseline specification ($\xi_1 = \xi_2 = 1$, diagonal Φ_λ and unrestricted K_λ), for $h = 0, \ldots, 10$ and $h = 1, \ldots, 10$ GFEVD shares. Impact connectedness is constant.